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ABSTRACT

Few-shot image generation remains challenging because limited training data can reduce image quality, diversity, and semantic consistency. In the Stable Attribute Group Editing (SAGE) framework, the use of the pixel2style2pixel (pSp) encoder can produce latent representations that lose local image details and limit editability during attribute group editing. This study proposes FeatureSAGE, an enhanced few-shot image generation framework that integrates the StyleFeatureEditor inverter and Feature Editor into SAGE. The proposed framework represents images using both W^+ latent codes and intermediate feature tensors, allowing semantic edits to be guided by latent-space directions while feature-level information supports detail preservation during synthesis.

FeatureSAGE was evaluated on the FUNIT Animal Faces and 102 Flowers datasets under a one-shot setting at 1024×1024 resolution. Quantitative evaluation used Fréchet Inception Distance (FID) to measure distributional alignment and Learned Perceptual Image Patch Similarity (LPIPS) to measure perceptual variation. On the 102 Flowers dataset, FeatureSAGE achieved an FID of 45.60, improving over SAGE by 15.8% and over AGE by 25.9%, with an LPIPS of 0.4209. On the Animal Faces dataset, FeatureSAGE achieved an FID of 49.60, improving over SAGE by 18.6% and over AGE by 20.2%, with an LPIPS of 0.4811. These results show that integrating feature-level editing with improved inversion strengthens generated-image distribution alignment while maintaining comparable perceptual variation. The study contributes a FeatureSAGE architecture that extends SAGE through StyleFeatureEditor-based inversion and feature editing for data-scarce generative modeling.

Keywords: Few-shot image generation, StyleGAN2, GAN inversion, attribute group editing, latent space

CHAPTER 1 INTRODUCTION

1.1 Background of the Study

Generative Adversarial Networks (GANs) have become a foundational framework for high-fidelity image generation. In a GAN, a generator and a discriminator are trained in a minimax game: the generator learns to produce synthetic images that fool the discriminator into classifying them as real, while the discriminator learns to distinguish generated images from real ones (Goodfellow et al., 2014). This adversarial process has enabled GANs to achieve remarkable realism in generated images (e.g. faces, animals) across many domains. In particular, the StyleGAN family of architectures (Karras et al., 2020) introduced a novel style-based generator that has set new standards for image quality. StyleGAN maps an input latent code z through a learned mapping network into an intermediate latent w in a space called W . Each layer of the generator is then modulated by w via adaptive instance normalization (AdaIN), and stochastic noise inputs add fine detail. This style-based design yields an intermediate latent space W that is significantly more disentangled than the original input space Z . As a result, controlled manipulation of individual semantic attributes (e.g. hair color, pose) is possible by adjusting w or the per-layer style codes. Subsequent improvements (StyleGAN2, StyleGAN3) refined the architecture and training, further boosting image fidelity and easing inversion (mapping real images back to latent codes).

StyleGAN’s latent representations come in several forms. The basic mapping from Z to W produces a single 512-D style code w per image. An extended latent space W^+ is also used, where a separate w vector is specified for each of the 18 synthesis layers. An even higher-dimensional feature space F can be formed by concatenating intermediate style and feature tensors (sometimes denoted F_k), capturing spatial details in the generator. Recent work has highlighted a trade-off across these spaces: moving from W to W^+ to F increases

reconstruction fidelity but reduces editability (Liu et al., 2023). In other words, encoding an image in the low-dimensional W space yields highly edit-friendly representations but may lose fine detail, whereas the high-dimensional W^+ or F spaces can reconstruct details at the cost of some entanglement and overfitting. StyleGAN2 in particular has shown that inversion in W^+ benefits from regularization (e.g. path-length regularization) to improve reconstructability and make projected images easier to attribute to latent codes (Karras et al., 2020).

To edit real images with a GAN, one must first perform GAN inversion: find a latent code that generates (nearly) the given image. This is a well-studied but challenging problem (Bobkov et al., 2024; Richardson et al., 2021). Early inversion methods used optimization (iteratively updating z to minimize reconstruction error), which can achieve high fidelity but is slow and may produce out-of-distribution latents. Encoder-based methods train a neural network to predict the latent directly, enabling real-time inversion at the expense of some quality. For StyleGAN, Richardson et al. (2021), proposed Pixel2Style2Pixel (pSp): a feed-forward encoder that maps any input image to a sequence of style vectors in W^+ (one w per layer). The pSp encoder can invert faces (or other images) into W^+ accurately without test-time optimization, greatly simplifying downstream editing. Generally, as noted by Richardson et al. (2021), “inversion methods either directly optimize the latent vector to minimize the error for the given image”. Encoder approaches like pSp offer fast inference, while optimization-based methods remain stronger on pure reconstruction quality. In either case, the goal of GAN inversion is to find an internal representation that contains as much image detail as possible and still allows semantic edits (Bobkov et al., 2024).

Once a real image is embedded in StyleGAN’s latent space, attribute editing becomes possible. It has been observed that semantic attributes (e.g. “smile,” “age,” “eyeglasses”) correspond to linear directions or subspaces in the latent space. For example, Shen et al. (2020) showed that well-trained StyleGAN latent codes encode a disentangled representation under linear transformations. In practice, one can move the latent w of an image along a learned direction Δw to change a particular attribute (e.g. $w + \alpha \Delta w$) (Ding et al., 2022).

Many works have exploited this: e.g. Linear SVM or PCA over latent codes can find directions for “smile” vs “no smile,” or for pose changes. This allows controlled editing via vector arithmetic. Importantly, style-based latents like W and W^+ are far less entangled than raw Z , so edits tend to be more semantically meaningful (Karras et al., 2020; Shen et al., 2020). Still, some entanglement remains: changing one attribute may inadvertently affect others. Recent methods (e.g. using subspace projection or network-based attribute classifiers) seek to further disentangle attributes. Moreover, higher-dimensional latent spaces (like W^+ or F) can encode finer identity details, enabling extremely high-quality reconstructions. Bobkov et al. (2024) leverage this by introducing a StyleFeatureEditor: an encoder that predicts both a W^+ code w and an intermediate feature tensor F_k . By editing in both w and F_k , they reconstruct and preserve finer image details even under heavy edits. In summary, StyleGAN latent spaces provide a powerful substrate for attribute editing, but the choice of W vs W^+ vs F presents a trade-off between edit stability and detail, and sophisticated encoders are needed to balance both.

A parallel line of research addresses few-shot image generation and editing. Unlike standard GAN training (which requires large datasets), few-shot generation aims to adapt a model to a new category or attributes given only a handful of examples. Meta-learning approaches have been applied here: for instance, the FIGR model Clouatre and Demers (2019) meta-trains a GAN with the Reptile algorithm so it can generate new classes from as few as 4 examples. Other works like FUNIT (Liu et al., 2019b) tackle few-shot domain translation, using image encoders to condition style generation on a few references. However, generating realistic variation from very limited data remains challenging. In particular, attribute group editing has emerged as a promising strategy for few-shot GANs. The idea is to collect many examples of known categories to learn generic attribute directions, then apply those directions to few-shot target categories. AGE learns a dictionary of “category-irrelevant” attribute vectors (e.g. pose, color) from seen classes and “category-relevant” class embeddings (mean latent per class) (Ding et al., 2022). At inference, given one or a few examples of an unseen category, AGE

embeds them into W^+ (using pSp) and manipulates those latents by combining the learned attribute directions. Similarly, Zhang et al. (2024) propose a Transformer-style approach with modules for learning discrete attribute codebooks and prompt-driven editing to handle diverse unseen classes. These methods show that by factorizing and reusing attribute vectors, a pre-trained GAN can generate plausible images of new categories without full retraining (Ding et al., 2022). Nonetheless, they often assume large amounts of labeled data for the seen categories and focus on one attribute at a time or on broad class attributes (like “breed of dog” vs “background color of bird”), rather than coordinated multi-attribute control.

Despite these advances, important gaps remain. Most GAN inversion and editing methods have focused on single-attribute manipulations or on scenarios with abundant data. Achieving stable, coordinated multi-attribute edits is still difficult, especially under few-shot conditions. For example, existing StyleGAN inversion methods either excel at reconstruction but produce entangled edits, or vice versa (Liu et al., 2023; Richardson et al., 2021). Moreover, according to Ding et al. (2023), “class-inconsistency” is a common problem GAN-generated images for downstream classification. Class inconsistency happens when the generative model corrupts some minor but critical information versus real pictures of a certain category (Ding et al., 2023). Few-shot models like AGE and TAGE demonstrate that group editing is possible, but they often do not explicitly ensure fidelity or identity preservation when combining many edits. Moreover, generation quality in few-shot setups can degrade due to mode collapse or identity drift. In face editing, for instance, we desire that an edited face remains clearly the same person (high identity similarity) while attributes like expression or hair style change. Conventional metrics like Fréchet Inception Distance (FID) and LPIPS capture distribution quality and perceptual similarity (Karras et al., 2020; Zhang et al., 2018b).

We proposed that achieving improved and stable group-wise attribute editing under few-shot conditions required an integrated approach. FeatureSAGE is designed to address this by combining replacing the standard pSp encoder of SAGE with StyleGANEditor Inverter and Feature Editor. The integrated encoder (inspired by StyleFeatureEditor) jointly predicts a W^+

latent w and a feature tensor F , aiming for high-fidelity inversion as well as explicit control of feature-level detail. This architecture is intended to preserve identity and image realism (as measured by identity similarity and low FID/LPIPS) while achieving disentangled, controlled modifications of target attributes. In summary, the background works on GANs, StyleGAN latent editing, and few-shot group editing together motivate FeatureSAGE’s goal: to provide stable, coordinated multi-attribute control in the latent space of a pre-trained GAN, even with only limited example images, without sacrificing image quality or attribute disentanglement.

1.2 Statement of the Problem

This study examines the main limitation of the current SAGE approach: its reliance on the pixel2style2pixel (pSp) encoder for GAN inversion. The pSp-based inversion stage can yield entangled latent codes and reduce control over semantic edits. In few-shot image generation, this limitation affects the production of semantically consistent and high-fidelity images from limited reference examples. Loss of local detail during inversion can also reduce the usefulness of generated images for downstream tasks because reconstructed images may no longer preserve the information present in the original samples.

The study therefore investigates whether improving the inversion and feature-editing stages of SAGE can strengthen both generated-image quality and editability. Specifically, the study seeks to answer the following research questions:

1. Can integrating the StyleFeatureEditor inverter and Feature Editor improve latent inversion fidelity and preserve image details for style-based editing within the SAGE framework?
2. Do these modifications influence the consistency, diversity, and quality of images generated in the few-shot setting, particularly for novel classes in the 102 Flowers and Animal Faces datasets?
3. How well does FeatureSAGE perform in terms of editing reliability and disentanglement

in few-shot scenarios?

4. How does FeatureSAGE compare with existing editing-based few-shot image generation methods, specifically AGE and SAGE, in terms of edit stability, category or identity consistency, and overall image quality?

1.3 Objectives of the Study

The objective of this study is to develop and evaluate FeatureSAGE, a few-shot image generation framework that improves upon SAGE by integrating the StyleFeatureEditor inverter and Feature Editor.

Specifically, this study aims to:

- a. Integrate the StyleFeatureEditor inverter and Feature Editor into the SAGE framework to improve generated-image quality through stronger reconstruction and feature-level editing.
- b. Evaluate the generated outputs on the Animal Faces and 102 Flowers datasets using FID and LPIPS.
- c. Benchmark FeatureSAGE against AGE and SAGE to examine improvements in edit stability, category or identity consistency, and overall image quality.

1.4 Significance of the Study

This study contributes to the few-shot image generation domain by enhancing the editability and interpretability of latent codes in the SAGE framework. By integrating the StyleFeatureEditor’s encoder and feature editor, the proposed FeatureSAGE framework improves stability and consistency in attribute group manipulation under data-scarce cases which addresses the key limitation of the existing few-shot generative models.

Aside from methodological contributions, the proposed approach also has practical relevance and has several downstream applications. In fine-grained visual recognition, FeatureSAGE can generate identity-preserving variations of an animal or plant from a very few dataset, which helps train classifiers for species or breed identification in environments where collecting large dataset is impossible or difficult to acquire. In biometric work that focuses on identity, such as forensic image analysis and identity-based data augmentation, FeatureSAGE can generate multiple edited versions of an image while keeping the same person's identity. This makes the findings more realistic and more reliable. In medical and scientific imaging, researchers often have small datasets because privacy rules limit sharing, and gathering new images can be expensive. In some cases, the condition or disease is rare. FeatureSAGE generates realistic images that can serve as a synthetic dataset for training and validating machine learning models.

This study extends prior work on attribute group editing and GAN latent spaces, and it gives clear guidance for building data-efficient image generation systems that can support downstream applications that need consistent, reliable image synthesis.

1.5 Scope and Limitation

This study is limited to few-shot image generation tasks using the StyleGAN2 generator. It specifically focuses on improving the reconstruction and editability of the model by integrating the StyleFeatureEditing Encoder and utilizing the Feature Editing Module into the framework of SAGE.

The datasets used in this study are restricted to Animal Faces and Flowers which offer diverse but structured categories suitable for evaluating attribute preservation and class coherence. The study does not explore optimization-based inversion techniques or apply to GAN architectures other than StyleGAN2. Results and conclusions are therefore constrained to the SAGE pipeline with these enhancements.

CHAPTER 2 REVIEW OF RELATED LITERATURE

2.1 Generative Adevrsarial Networks

Generative Adversarial Networks (GANs) is a novel framework for training generative models via an adversarial process which was first introduced by Goodfellow et al. (2014). In the paper, "Generative Adversarial Nets", a generator network G produces samples aimed at mimicking the data distribution, while the discriminator D estimates the probability of whether an input is real or generated (Goodfellow et al., 2014). In other words, GAN framework basically corresponds to a minimax two-player game where, a generator (G) creates samples that are intended to come from the same distribution as the training data, while the discriminator D examines the probability of the the samples provided by G being real or fake (Goodfellow, 2017).

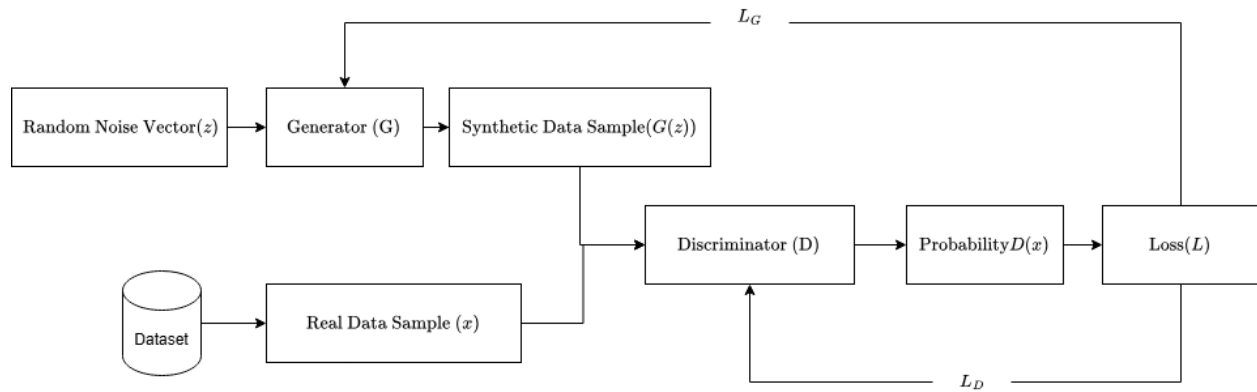


Figure 2.1 Generative Adversarial Networks (GANs) Architecture

As shown in Figure 2.1 the generator G takes as input a random noise vector $z \sim p_z$ (usually drawn from a simple distribution such as uniform or Gaussian) and outputs a synthetic data sample $G(z)$. The goal of G is to map the "latent" noise distribution into the data sample $G(z)$ is as realistic as possible. The discriminator D takes a data sample x as input and outputs a probability $D(x)$ estimating the likelihood that x is a real sample rather than one produced by

G (Salehi et al., 2020; Cohen and Giryas, 2022; Creswell et al., 2018).

The adversarial training process in GANs is governed by a minimax optimization. Formally, Goodfellow et al. (2014) define the value function:

$$V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))]$$

and train D and G in a two-player game:

$$\min_G \max_D V(D, G)$$

D and G are alternatively updated via minibatch stochastic gradient descent during the GAN training process. D is trained to maximize the probability of predicting whether an input from both training example and sample from G are real or fake, while G is trained simultaneously to minimize $\log(1 - D(G(z)))$ (Goodfellow et al., 2014). In reality, GAN training can be delicate: if G becomes too strong or D too weak, gradients can vanish, and if D is too strong early on, G may receive no learning signal (saturating loss). Goodfellow et al. (2014) suggest a non-saturating heuristic for G (maximizing $\log D(G(z))$) instead of minimizing $\log(1 - D(G(z)))$ to provide stronger gradients in practice.

On its first introduction, GANs have brought about advantages compared to other generative models at the time (e.g. Variational Auto-encoders(VAE)) like sharper images, configurable sizes, richer search space, and a veratility as it can support different generator networks. However it also has its disadvantages like : mode collapse, vanishing gradients, and instability (Cohen and Giryas, 2022).

In the years following the initial discovery of Generative Adversarial Networks (GANs), there have been numerous improvements made to their architecture that have resulted in better image quality, more stable training, and expanded capabilities. Early research highlighted the importance of convolutional structures for successful GAN performance. One influential contribution was made by Richardson et al. (2021), who introduced the Deep Convolutional

GAN (DCGAN) and a set of practical design recommendations for building effective convolutional GANs. Instead of employing traditional pooling or unpooling layers, DCGAN utilized strided convolutions to decrease spatial resolution and transposed convolutions to increase it. This allowed both G and D to learn spatial features more effectively and led to outputs that were more realistic (Goodfellow, 2017). They also recommended batch normalization in almost all layers of both G and D , excluding the output and input layers respectively, which greatly stabilized training and facilitated learning proper scales.

These guidelines enabled DCGANs to synthesize high-resolution images in “one shot” and learn disentangled latent representations that enable simple vector arithmetic for changing semantic attributes of the output. DCGAN also popularized the use of Adam as an optimizer for GANs (Goodfellow, 2017). In conclusion, this research provides a comprehensive overview of how convolutional structures can improve the performance of GANs, and how practical design recommendations such as strided convolutions, transposed convolutions, batch normalization, and Adam can enable these improvements. By following these guidelines, it is possible to build high-quality image synthesis models that learn complex spatial relationships in images and produce realistic outputs. In particular, DCGANs are able to synthesize high-resolution images in “one shot” and learn disentangled latent representations that allow for simple vector arithmetic for changing semantic attributes in the output, as well as enable more flexible model configurations by allowing the use of a different optimizer, such as Adam, which has been shown to be effective in training GANs. Furthermore, DCGANs popularized the use of batch normalization across all layers in both the generator and discriminator networks, excluding the output and input layers respectively, which greatly stabilized training and facilitated learning proper scales. These findings highlight the importance of convolutional structures in improving the performance of GANs and provide a comprehensive overview of how practical design recommendations can be used to build high-quality image synthesis models that learn complex spatial relationships in images and produce realistic outputs.

While StyleGAN and StyleGAN2 have achieved impressive results for image realism, they

still face some challenges. The main limitation of StyleGAN lies in its inability to accurately imitate real-world images and in the fact that it cannot directly manipulate the latent space within the network. Additionally, even when inversion is implemented, the resulting images may not always capture identity or be structurally accurate. Despite these limitations, StyleGAN has shown promise in some unstructured scenes and faces. However, as Bermano et al. (2022) observed, StyleGAN struggles with domains that do not exhibit strong structure and may even replicate demographic biases in the training set. Finally, while StyleGAN can generate images at high resolution with ease, it requires extensive GPU resources to train on high-resolution data. Moreover, careful tuning of the architecture is also necessary to ensure fairness and accuracy. Overall, while StyleGAN has achieved significant progress for image realism, there are still challenges that need to be addressed in order to fully utilize its potential for creative and sophisticated use cases.

2.2 StyleGAN and StyleGAN2

Early GANs suffered from instability and mode collapse, which Karras et al. (2018) addressed by Progressive GANs (PGGAN): training begins at low resolution and gradually adds layers as resolution increases (Melnik et al., 2024). Building on this, StyleGAN Karras et al. (2019) proposed a novel style-based generator architecture that achieved state-of-the-art image quality and disentangled latent factors. StyleGAN is a variant of GAN that builds on its framework by radically redesigning the generator.

Instead of feeding the latent code $\mathbf{z}$ directly into the generator, StyleGAN first maps $\mathbf{z} \sim \mathcal{N}(0, I)$ to an intermediate latent $\mathbf{w} = f(\mathbf{z})$ using an 8-layer fully-connected network (Karras et al., 2019). This $\mathbf{w}$ lives in a new “style” space $\mathcal{W}$ that is learned to be less entangled than the original latent space. Learned affine transforms then convert $\mathbf{w}$ into per-layer style parameters (y_s, y_b) that modulate each convolution via Adaptive Instance Normalization (AdaIN) (Karras et al., 2019, 2020). Concretely, for each feature map x_i in a convolutional layer, StyleGAN

applies:

$$\text{AdaIN}(x_i, y) = y_{s,i} \left(\frac{x_i - \mu(x_i)}{\sigma(x_i)} \right) + y_{b,i}$$

, where y_s and y_b (scalars for channel i) are derived from $\mathbf{w}$. In practice, StyleGAN’s mapping network f (an eight-layer MLP) produces $\mathbf{w}$, and then each convolutional layer is modulated by the corresponding (y_s, y_b) pair. This allows scale-specific control of features: coarse spatial styles (e.g. pose or overall shape) are set by early layers, while fine-grained details (e.g. color, texture) are set by later layers Melnik et al. (2024); Karras et al. (2019). Notably, the mapping network and style modulation enable style-mixing: two latent codes can be injected at different depths, blending coarse features from one code with fine details from another (Melnik et al., 2024).

Importantly, StyleGAN replaces the standard random input with a learned constant $4 \times 4 \times 512$ tensor, and all variation comes from the style vectors and added noise. At each convolutional layer, StyleGAN also injects a random noise map (one per pixel) which is added to the features with learned per-channel weights. This stochastic noise produces high-frequency variation (e.g. hair wisps, freckles) without affecting global attributes. Together, these design choices yield unsupervised disentanglement of high-level attributes from stochastic detail: Karras et al. (2019) observe that their style-based generator “leads to an automatically learned, unsupervised separation of high-level attributes (e.g., pose and identity) and stochastic variation in the generated images”. In experiments, StyleGAN achieves state-of-the-art image fidelity (low Fréchet Inception Distance) and more linear, less entangled latent factors than conventional GANs.

The architectural differences from a standard GAN can be summarized as:

- Intermediate latent space and mapping network: StyleGAN learns a mapping $f : \mathbf{z} \rightarrow \mathbf{w}$ (eight-layer MLP) so that $\mathbf{w}$ in space $\mathcal{W}$ is disentangled
- Style modulation via AdaIN: Style vectors from $\mathbf{w}$ provide per-layer scale/bias that modulate each convolution by AdaIN, as in the formula above.

- Noise injection: Learned Gaussian noise is added to each layer to control stochastic details
- Learned constant input: The generator begins from a fixed learned 4x4 tensor instead of random noise
- Progressive growing: StyleGAN adopts the progressive training strategy from ProGAN [Karras et al., 2018], starting at 4x4 resolution and gradually adding layers to reach high resolution (Karras et al., 2018).

StyleGAN2 Karras et al. (2020) builds directly on StyleGAN but redesigns the generator to eliminate artifacts and boost quality. The key insight is that the original AdaIN normalization can destroy meaningful signal: normalizing each feature map breaks global correlations needed for coherent details. Therefore, StyleGAN2 “bakes” the style modulation into the convolution weights and replaces the per-feature normalization with a weight demodulation operation. In practice, this works as follows: let w_{ijk} be the convolution weight connecting input channel i to output channel j (at spatial position k). Given a style scale s_i (from $\mathbf{w}$) for channel i , StyleGAN2 first multiplies w_{ijk} by s_i (modulation), yielding $w'_{ijk} = s_i w_{ijk}$. It then computes the output standard deviation σ_j of the convolution and demodulates by dividing by σ_j :

$$w''_{i,s,j,k} = \frac{w'_{ijk}}{\sqrt{\sum_{i'} (w'_{i'jk})^2 + \epsilon}}$$

. This eliminates the need for explicit instance normalization after the convolution. The result is that each convolutional layer can apply the style s_i directly to its weights, with the demodulation automatically preserving unit variance. This modification removes the “blob” artifacts common in StyleGAN outputs and leads to crisper, more natural images.

Training in StyleGAN2 takes a different path. Instead of slowly increasing complexity, the model uses a set structure right away. Skip connections and residual blocks link features across scales, much like in MSG-GAN or U-Net designs. This setup keeps signal flow strong throughout

training. Without needing to add layers over time, stability improves naturally. The approach skips the extra burden of gradual expansion. Yet, key elements remain - style mapping and noise inputs stay part of the process. These are simply embedded into the updated generator framework. Stability meets continuity here. Not every change means replacing what worked. Another finding from Karras et al. (2020) shows that applying path length regularization to the mapping network helps balance how shifts in W affect visual output. Because of this adjustment, movements through latent space translate more consistently into noticeable image differences. The result is a smoother connection between codes and images, easier reversal from image back to code, better separation of features, along with improved editing control.

The architectural improvements in StyleGAN2 can be summarized as:

- Weight demodulation instead of AdaIN: Styles modulate convolution weights and are normalized in-weight, removing all explicit feature-map normalization.
- No instance normalization layers: By design, the generator no longer uses AdaIN or other instance/batch norms; all normalization is implicit in weight demodulation
- Elimination of progressive growing: The network is trained at fixed full resolution, using skip/residual connections to preserve stability
- Path length regularization: A regularizer on the mapping network encourages a smooth, linearized latent space.

These changes yield a substantial quality gain. For example, Karras et al. (2019) report that fully updated StyleGAN2 (no progressive growing, weight demodulation, path regularization) achieves a Fréchet Inception Distance of about 3.3 on FFHQ, compared to 4.4 for the original StyleGAN baseline.

Though StyleGAN and its successor deliver remarkable visual fidelity, certain constraints remain evident. Generated outputs along with manipulation options depend strictly on what

the model learned during training. Control through latent space works solely within the boundaries of synthetic imagery the system supports; actual photographs require conversion into domain via GAN inversion prior to adjustments (Bermano et al., 2022). Yet perfect preservation of subject traits isn't always achieved post-inversion. What makes face images easier for these models becomes a hurdle elsewhere - disorganized settings confuse their output. Bermano and team noted back in 2022 how StyleGAN falters when patterns lack clear repetition. Problems show up beyond performance too, since learned behaviors include social imbalances present in source data. Training runs expose another layer of difficulty: high-res workloads need powerful graphics processors, precise adjustments. Complexity piles up, even if results look effortless.

2.3 The Latent Space

Early GANs drew a latent vector z from a simple distribution and fed it through a generator, but the resulting latent space was largely entangled and hard to control. InfoGAN by Chen et al. (2016) introduced an information-maximizing objective to encourage disentanglement, yielding interpretable latent factors. The progressive GAN (PGGAN) of Karras et al. (2018) improved image quality via multiresolution training but still used a single latent code at the input. A major shift occurred with StyleGAN Karras et al. (2019), which used a learned mapping network to transform the input z into an intermediate latent W space. This mapping, via 8 fully-connected layers, “unfolds” the simple z distribution into W , aligning it to the generator's feature distribution and enabling axis-parallel edits (Melnik et al., 2024). StyleGAN also injects this W code and a random per-layer noise at each convolutional layer, allowing layer-wise control of coarse-to-fine image attributes.

StyleGAN2 built on this by replacing Adaptive Instance Normalization (AdaIN) with weight modulation and demodulation; it further regularized the latent mapping to smooth out the generator's response to latent shifts (Karras et al., 2020). This made the latent-to-image mapping more uniform and significantly easier to invert, meaning real images could

be projected more reliably into W . Importantly, StyleGAN2 popularized the use of an extended latent space W^+ , in which a separate 512-dim w vector is supplied to each of the 18 generator layers. In W^+ , each layer can have its own style code, greatly increasing representational power: for a 1024x1024 model, W^+ is 18x512-dim. However, W^+ vectors do not come from the learned W distribution, so naive sampling can produce realistic “face-like” images but also novel patterns that are still face-like (eg. Aliens) (Melnik et al., 2024). W^+ is used mostly for reconstruction/inversion and fine edits, whereas W remains the base latent space for smooth, in-distribution sampling. Finally, StyleGAN2’s style code w is further transformed, once per layer, into a channel-wise affine vector s in Style Space S , which directly scales convolutional feature maps (Melnik et al., 2024). In StyleGAN2, each channel’s style parameter in S is a single scalar, controlling only scale (no bias). Thus StyleGAN2 introduced multiple nested latent spaces: the input Z , the mapped W , the layered W^+ , and the channel-wise S .

The StyleGAN architecture encourages disentanglement by design. Karras et al. (2020) observed that W becomes “much closer to the required feature distribution” than z , so that semantic edits align with simple directions. In StyleGAN the W space yields more disentangled representations than the original Z space, making long-range manipulations (e.g. aging) more robust (Shen et al., 2020). InterFaceGAN explicitly exploits this: it identifies linear hyperplanes in W corresponding to binary face attributes (smile, age, eyeglasses, etc.) and shows that moving w across these decision boundaries toggles the attribute. In StyleGAN’s W space, these attributes become nearly disentangled after a linear transformation, unlike in Z .

Unsupervised methods have since discovered many such directions. GANSpace applies Principal Component Analysis (PCA) to StyleGAN2 latent codes and finds that principal components correspond to high-level edits (pose, color, environment) (Harkonen et al., 2020). Remarkably, GANSpace shows that a single global PCA direction, when applied at specific layers (“layer-wise decomposition”), can control attributes from coarse (camera angle) to fine (facial features) without labels. Similarly, Voynov and Babenko (2020) developed an unsupervised optimization to extract interpretable latent directions from any GAN; they

report non-trivial findings (e.g. a “background removal” direction) without any supervision. Semantic Factorization (SeFa) by Shen and Zhou (2021) provides a complementary approach by performing a closed-form eigen-decomposition of the pretrained generator’s weight matrices, SeFa factors the W space into semantically meaningful directions (necklace on/off, lighting, etc.) without training or samples. Empirically, SeFa’s directions often match or exceed those from supervised methods (e.g. InterFaceGAN) in manipulating specific attributes.

The disentangled latent structure of StyleGAN2 enables rich image manipulation. A common paradigm is latent traversal: given a latent code w , one perturbs it along a semantic direction and observes the image change. This can be simple as linear arithmetic like how InterfaceGAN uses $z_{edit} = z + \alpha n$ for single attribute manipulation. More advanced controllers have been developed. StyleFlow (Abdal et al., 2021) models latent editing as a continuous conditional flow: it learns a bijective mapping in latent space conditioned on attribute values (age, pose, gender, etc.), enabling smooth attribute-conditioned sampling and editing (Abdal et al., 2021). In StyleFlow, one can fix most aspects of the face and continuously change a specified attribute like gradually make the person older with high fidelity. Similarly, StyleCLIP leverages CLIP text embeddings to find latent directions: it either optimizes a latent to match a text prompt or trains a mapper network to translate text into a direction in W or S . This yields a flexible text-based interface for latent edits (“smile”, “pikachu”, “gothic”) without requiring explicit labeled data. (Patashnik et al., 2021).

For image editing tasks, one often needs to map a real photo into StyleGAN2’s latent space. GAN inversion methods seek a latent code (in W , W^+ , or even feature space F) that reproduces the input under the generator. According to Liu et al. (2023), GAN-Inversion methods suffer from fidelity-editability trade-off where optimizing in W^+ (high capacity) yields near-perfect reconstruction, but the resulting w^+ may lie outside the “safe” distribution, making edits unpredictable. Conversely, constraining to W yields more stable edits but higher reconstruction error. Modern techniques address this: encoder-based methods like pixel2Style2pixel (pSp) (Richardson et al., 2021) or encoder4Editing (e4e) (Tov et al., 2021)

quickly predict an approximate W^+ code; then post-processing (latent optimization or a “pivotal tuning”) refines it. For example, pSp’s encoder directly outputs 18 layer-specific style vectors into W^+ , inverting a real face in one forward pass. Such hybrid inversion plus fine-tuning allows one to start with an interpretable latent in W/W^+ and then apply the semantic edits learned for synthetic images.

Ultimately, StyleGAN2’s latent space serves as a controllable image model. One can generate images with specified attributes by sampling z and then steering $w = M(z)$ in desired directions (Patashnik et al., 2021). For instance, adding a “smile” direction to w yields smiling faces. More generally, the latent allows fine-grained tuning: by selecting w^+ values per layer, one can combine styles from different images or enforce multi-attribute conditions. StyleGAN2’s latent space, especially W and S , provides a semantically meaningful coordinate system. Its evolution from earlier GAN latents (simple Z) to this multi-space structure underpins the state-of-the-art in GAN-based image editing and disentangled generation.

2.4 Attribute Editing in GAN Latent Spaces

Modern Generative Adversarial Networks (GANs) have shown that the latent codes controlling image generation often embed rich semantic structure. In particular, many high-level attributes (e.g. age, smile, pose) correspond to almost-linear directions in the latent space (Shen et al., 2020; Harkonen et al., 2020). Once a direction vector d for an attribute is found, editing is done by simple latent arithmetic $w_{edit} = w + \alpha d$ where w is the original latent code α is a scalar step size, and d is the learned attribute direction. For example, InterFaceGAN treats attributes as binary labels and trains a linear SVM in the latent space; the normal vector of the separating hyperplane becomes d (Shen et al., 2020). Style-based GANs (StyleGAN1/2) enhance this by using a mapping network to produce an intermediate latent W space that is empirically more disentangled than the input Z space (Karras et al., 2020; Wu et al., 2020).

There are two main approaches to finding attribute directions. Supervised methods use labeled data: for a given attribute one can train a linear classifier (e.g. on whether images

have glasses) on a set of latent vectors. The classifier’s weight vector (normal to the decision boundary) becomes the edit direction. InterFaceGAN demonstrated this works for many facial attributes (Shen et al., 2020). There are two main approaches to finding attribute directions. Supervised methods use labeled data: for a given attribute one can train a linear classifier (e.g. on whether images have glasses) on a set of latent vectors. The classifier’s weight vector (normal to the decision boundary) becomes the edit direction. InterFaceGAN demonstrated this works for many facial attributes. Unsupervised methods do not require labels: GANSpace, for instance, collects many random latents and applies Principal Component Analysis to find dominant directions. These principal axes often correspond to pose, lighting, or object type. Other factorization techniques, like SeFa, and style-space channel-searchers similarly find promising axes with little or no supervision (Shen et al., 2020; Harkonen et al., 2020; Shen and Zhou, 2021).

Once a direction d is chosen, editing is simply walking along that direction in latent space. Concretely, after inverting or sampling a code w , one produces an edited image by generating from $w + \alpha d$ for various α . InterFaceGAN likewise shows that adding the gender vector shifts a face’s apparent gender. In general, Attribute Editing Models use the same principle of finding d or Δw and setting $w_{edit} = w + \alpha d$ to change attributes in an image (Harkonen et al., 2020; Shen et al., 2020; Li et al., 2023; Wu et al., 2020)

Importantly, these methods rely on latent disentanglement: the idea that different factors are (at least partly) separable. StyleGAN’s design helps here: its intermediate latent W is empirically more disentangled than the raw Z space (Wu et al., 2020). StyleGAN2’s per-channel style parameters (“StyleSpace”) have been shown to control very localized attributes (e.g. hair color, background elements) with minimal interference. In contrast, earlier GANs had more entangled latents, making clean edits harder (Salehi et al., 2020; Creswell et al., 2018).

However, disentanglement is not perfect. Most discovered directions still have residual cross-attribute effects. For example, one edit vector might change both “smile” and “age” together. This entanglement is noted in the literature: many control directions affect multiple

attributes and can be “non-local” (Wu et al., 2020). To mitigate this, some methods perform conditional manipulation: e.g. subtracting out the projection of one direction on another to isolate each effect. But instability remains when editing multiple attributes at once, often producing unrealistic images if not handled carefully.

Attribute editing assumes that one has latent code w to edit . To apply editing to real photos, a critical step is GAN inversions. By finding a latent code that faithfully reproduces a given real image, one can apply learned semantic manipulations ,adjusting age, hairstyle, or pose, to that image (Bobkov et al., 2024; Alaluf et al., 2021). We find a latent code z^* whose generator output matches the target image. Once z^* is found , one can apply the w_{edit} formula to manipulate the image attributes. According to Xia et al. (2022), there 2 gan inversion methods. Namely, the learning-based(encoder-based) and optimization based Gan inversions.

- Optimization-based inversion solves for z^* by backpropagation: it iteratively adjusts z to minimize reconstruction loss $\|G(z) - x\|$ This often yields high-fidelity reconstructions but can be slow and may get stuck in local minima.
- Learning(Encoder)-based trains a neural network $E(\cdot)$ to map images to latent codes. A trained encoder can invert images quickly, though it must be carefully regularized to align with the pretrained generator. Hybrid approaches use an encoder for an initial guess followed by optimization refinement (Xia et al., 2022).

GAN inversion is what allows editing of real images. For instance, the survey by Xia et al. (2022) illustrates how an inverted code z^* can be altered by adding attribute vectors n_1, n_2 to change the age or smile of a real face. They note that after inversion, the same latent-space editing techniques apply “without requiring any ad-hoc supervision” However , inversion quality has limits: early models like Progressively Growing Gan (PGGAN)(Karras et al., 2018) were hard to invert faithfully, whereas StyleGAN’s architecture yields much better inversions (Karras et al., 2020). Imperfect inversion can blur details or introduce artifacts, which in turn limits how well edits carry over to the final image.

2.4.1 Attribute Group Editing

Beyond single images, recent work uses latent editing for few-shot image generation: synthesizing new images of a novel category given only a few examples. The idea is to treat each new class as a 'set of attributes' and apply the learned edit directions to produce more samples. A pioneering approach is Attribute Group Editing (AGE) (Ding et al., 2022). AGE assumes that images are combinations of category-relevant and category-irrelevant attributes. It represents a category by its class embedding w_c : the mean latent of the few examples. The attributes that define the category (e.g., animal shape) are captured in w_c . AGE then learns a sparse dictionary of category-irrelevant attribute directions (e.g. fur color, pose) by doing sparse coding on the differences $w_{sample} - w_c$ for many samples. The key assumption is that an attribute direction (such as 'smiling' or 'red flower') is shared between categories. To generate new images of an unseen category, AGE starts with the class embedding w_c and adds combinations of these dictionary directions, creating diverse variations while keeping the features of the core category fixed. This editing-based generation requires no GAN retraining and can produce high-quality, diverse images even from one or few inputs.

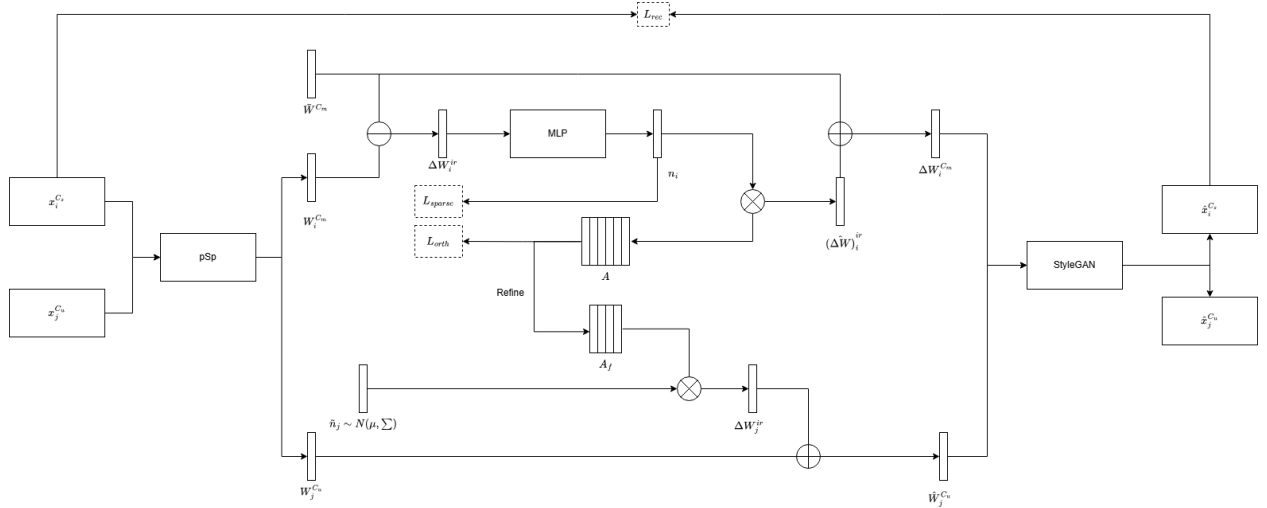


Figure 2.2 Attribute Group Editing Framework

AGE provides a powerful framework, but has some limitations. The SAGE (Stable AGE) follow-up aims to improve class consistency. Instead of editing from just one example, SAGE

uses all a few-shot images to robustly estimate the class embedding. It leverages the same attribute dictionaries but selects appropriate directions based on the new class’s similarity to the seen classes. In effect, SAGE ‘injects the whole distribution of the novel class into StyleGAN’s latent space’, preserving the identity of the category. This leads to more stable outputs that help downstream tasks (e.g., classification) perform better on the generated images (Ding et al., 2023).

Most recently, TAGE (Trustworthy AGE) has extended this line further. TAGE also uses a codebook (dictionary) of attributes, but introduces transformer-style modules. Its Codebook Learning Module (CLM) finds semantic directions for category-related and unrelated attributes and builds a sparse dictionary. A Code Prediction Module (CPM) then uses global combinatorial reasoning (long-range attention) to predict the best edit codes for a given few-shot class, improving diversity and accuracy. Crucially, TAGE adds a Prompt-Driven Semantic Module (PSM): it generates textual or semantic prompts that are injected into transformer layers, guiding fine-grained editing (Zhang et al., 2024).

2.4.2 Stable Attribute Group Editing

Stable Attribute Group Editing (SAGE) was proposed as a method to overcome such limitations of AGE. Instead of editing from a single exemplar, SAGE initially employs all the given few-shot images to estimate a stable class-center embedding. It does so by capitalizing on the pretrained category-relevant attribute dictionary (learned from visible classes) to linearly represent the novel class centroid in the latent space. As described by Ding et al., since there are not a sufficient number of examples to calculate a credible mean directly, “the category-relevant dictionary... is complete enough to cover the unique attributes of any related categories,” a linear combination of the dictionary atoms can “recover the class-specific features of the unseen classes” (Ding et al., 2023).

In practice, SAGE projects the few-shot samples onto such an attribute dictionary to separate their shared class features from each other, essentially removing the attributes not

related to categories and retaining the attributes that are related. According to the authors, such a disentanglement also helps the model to estimate a more accurate class embedding even for a very small number of samples. The experiments also indicate that the estimated embedding from only three face images well approximates the ground truth centroid over hundreds of actual instances.

In the same paper, the weights from the projection are utilized for adaptively choosing which category-irrelevant editing directions to use. Taking cues from biological taxonomy, the authors make the assumption that categories sharing the same relevant characteristics which is being members of the same genus, for example they also share compatible and safe irrelevant characteristics. Consequently, SAGE picks the category-irrelevant directions from the highly similar seen classes depending on the projection coefficients. The approach enables the method to introduce the distribution of the unseen class to StyleGAN’s latent space by essentially repositioning the generated sample cloud towards the actual class manifold. Anchoring the synthesis about this reconstructed embedding and then carrying out filtered edits, SAGE manages to maintain class identity despite generating diverse and realistic variations.

This refinement of AGE yields clearer gains in category consistency and diversity. Quantitative results show that SAGE produces samples with significantly better quality (lower FID) and greater diversity (higher LPIPS) than AGE and competing few-shot methods. For example, without retraining the GAN, “SAGE generates images with higher quality and diversity” than prior fusion- or transformation-based approaches, and its class-embedding mapping makes it “much more stable than AGE”. Qualitatively, Ding et al. illustrate that AGE could produce bizarre artifacts (a “cat with a dog nose” or a “bird with two heads”) when editing from extreme inputs, whereas SAGE’s tailored embedding and adaptive edits yield realistic variants without such errors. In the latent space of an unseen class, AGE’s few-shot samples often lie at extreme positions and its generated points cluster narrowly around them. By contrast, SAGE’s injections form a much wider latent cloud that more closely matches the true distribution. The authors summarize that SAGE “relocates the whole distribution of the novel class in the latent

space, thus effectively avoiding category-specific editing and enhancing the stability”. Overall, Ding et al. report that SAGE achieves “more stable and reliable few-shot image generation” with significantly better category retention than previous methods.

Nonetheless, SAGE does not solve all challenges. The authors note that its generative range is still more restricted than that of real data: the “dynamic range” of editable variations in SAGE is smaller than in the true class. In practice, SAGE currently uses a uniform editing intensity for all directions (since estimating per-attribute scales from few samples is infeasible) and its dictionary contains only attributes that recur across the seen classes. Consequently, some subtle class-specific variations may still be missing. As the paper observes, “SAGE simply assumes a unified intensity for editing direction sampling,” and capturing attribute-dependent editing amplitudes or class-specific edits remains an open issue. Moreover, while classification performance is improved over AGE, it still generally lags behind a model trained on real examples. This gap – and the reliance on a pretrained StyleGAN inversion as the generator – highlight future work: extending SAGE’s ideas to handle multi-scale attributes and other generative models may be needed to fully close the gap for few-shot tasks.

2.4.3 Datasets and Applications

Attribute editing techniques are typically developed and evaluated on diverse image domains. The datasets Animal Faces HQ and Flowers are common: they have many labeled attributes and StyleGAN models are readily available for them. Few-shot generation research has also expanded to other domains. For instance, the Flowers dataset has 102 flower species (Gupta et al., 2024), and Animal Faces (or AnimalFacesHQ) has 149 animal species.

Latent editing has practical use in data augmentation (few-shot generation) and interactive image manipulation. For example, one can start with a single animal or flower image and synthesize variants by editing color, pose, or environment using learned directions. However, note that many attributes do not directly map across domains; cross-domain transfer of semantic concepts remains challenging.

2.4.4 Attribute Editing limitations and challenges

Despite their promise, latent-space attribute editing methods face several challenges (Wu et al., 2020; Ding et al., 2022, 2023; Shen et al., 2020; Xia et al., 2022; Liu et al., 2023):

- **Attribute entanglement.** Discovered directions often change multiple features at once. For instance, moving along an “age” vector might also alter “expression”. Entangled edits require careful design (e.g. subtracting projections) or more sophisticated disentanglement techniques. Fully disentangling a GAN’s latent space is still an open problem.
- **Multi-attribute interactions.** Applying more than one edit vector can be unstable. Sequential edits may accumulate artifacts, and combining directions (e.g. “add glasses” and “increase smile”) can produce unrealistic results if the attributes interact in complex ways. Ensuring consistent multi-attribute editing (for example, editing several facial features simultaneously without distortion) remains tricky.
- **Inversion quality.** Editing real images relies on GAN inversion. If the inversion does not perfectly capture the original, edits may fail or degrade fidelity. Early GANs (like PGGAN) struggled with inversion. While StyleGAN produces better inverted results, truly lossless inversion is hard. Any residual reconstruction error limits how cleanly attributes can be edited.
- **Generalization to diverse datasets.** Most latent-editing research has centered on well-studied domains (faces, common objects). Extending these methods to fine-grained categories like birds or flowers raises issues. The learned GAN (or its latent semantics) may not capture useful directions in these domains, especially if attribute labels are scarce. Additionally, attributes meaningful in one domain (e.g. “nose shape” in faces) do not translate to others. New methods must learn or adapt directions that make sense for non-face classes.

2.5 GAN Inversion

GAN inversion refers to finding a latent code in a pretrained GAN (here StyleGAN2) that reproduces a given real image when passed through the generator. In practice, one inverts an image x by solving for a style code w such that the StyleGAN2 generator G outputs an image $G(w) \approx x$ (Tov et al., 2021). This can be done via optimization (iteratively adjusting w for each image, which is slow) or via a learned encoder (a neural network that predicts w in one forward pass) (Alaluf et al., 2021; Tov et al., 2021). StyleGAN2 uses a mapping network from a base Gaussian z into an intermediate latent space W , which is more disentangled. Many inversion methods actually use an extended latent space W^+ (one 512-D style vector per layer) for greater expressiveness (Richardson et al., 2021; Tov et al., 2021; Bobkov et al., 2024). Once an image is inverted to a latent code, one can edit the image by manipulating that code (e.g. adding learned attribute directions or swapping style layers) and generating the edited image via G . High-quality inversion thus requires balancing reconstruction fidelity (low distortion between x and $G(w)$) with editability (the code lies in a region where semantic edits are meaningful).

How GAN inversion works (Richardson et al., 2021; Alaluf et al., 2021; Tov et al., 2021):

- **Pretrained StyleGAN2:** Use a generator G trained on a target domain. Fix its weights.
- **Latent spaces:** Identify the latent space to invert into (typically W or extended W^+ sometimes an additional feature space F).
- **Encoder or optimization:** For encoder-based inversion, train a feed-forward network E to map images to latents. For optimization-based inversion, for each image x optimize W by minimizing a loss $\|x - G(w)\|$. Encoder methods are much faster (one pass) but historically had worse fidelity than optimization.
- **Architecture of encoders:** Typically, an encoder is a convolutional backbone (often ResNet) that produces latent parameters. For example, pixel2style2pixel (pSp) uses a

ResNet + feature pyramid, producing style vectors via map2style blocks for each scale. Encoder4Editing (e4e) is similar but outputs one base style plus per-layer offsets. ReStyle applies such encoders iteratively, and StyleFeatureEditor uses a two-branch encoder for both W^+ and feature-latents.

- **Loss functions:** Encoders are trained to minimize a combination of losses on reconstruction. Common losses are pixel-wise L2, perceptual (LPIPS) loss, and often an identity or feature loss to preserve semantics (e.g. using ArcFace or self-supervised features). Many methods also add regularization: for example, e4e adds L2 penalties on offsets to keep the final code near W , and some use adversarial (latent discriminator) loss to keep codes within the true latent distribution.
- **Inference and editing:** At test time, given image x , the encoder predicts a latent code (or code+offsets or code+features). This code is fed into the fixed generator G to reconstruct x . For editing, one modifies the latent (e.g. by style mixing or adding semantic direction vectors) and uses G to synthesize the edited image.

2.5.1 pixel2style2pixel (pSp)

pSp is an early StyleGAN2 encoder (CVPR 2021) that maps an image to the extended latent space W^+ . It uses a ResNet-50 backbone with a feature pyramid producing three scales of feature maps. Each scale is fed through small map2style networks to generate 512-D style vectors corresponding to StyleGAN’s layers. The output is a full W^+ . The encoder is trained with a weighted sum of losses: pixel L2, LPIPS perceptual loss, and an identity loss (ArcFace for faces, or a self-supervised ResNet for other domains). A small regularization term on the styles can also be used. By design, pSp prioritizes reconstruction fidelity. It “directly maps a real image into the W^+ latent space with no optimization required”. In experiments it achieves low LPIPS and MSE reconstruction error. Because it predicts full W^+ , pSp can accurately capture fine details of the image. After encoding, image editing is done by standard StyleGAN manipulations (e.g. adding latent offsets or style-mixing). For example, pSp naturally supports

style-mixing for multi-modal synthesis: one can replace selected style vectors with those from a random code to generate variations.

pSp is fast (single forward pass) and yields high-quality reconstructions. It has been shown to preserve identity and details (e.g. hairstyle, lighting) better than many earlier methods. It also generalizes to various image-to-image tasks without retraining (e.g. frontalization, super-resolution) because of the rich StyleGAN latent space.

However, since pSp uses full W^+ , the encoded latents can lie far from the original distribution learned in W . This can degrade “editability”: edits in regions of W^+ far from the data manifold may produce unrealistic results. pSp’s very high reconstruction fidelity sometimes comes at the cost of less coherent edits, compared to encoding into W or constrained spaces. (Richardson et al., 2021)

2.5.2 Encoder4Editing (e4e)

Encoder4Editing (e4e, ICCV 2021) extends pSp by explicitly addressing the reconstruction editability trade-off. The key idea is to keep the inverted latent closer to the original W distribution even while allowing high-fidelity reconstruction. Architecturally, e4e still uses a ResNet+feature pyramid encoder, but instead of predicting all style vectors independently, it predicts one base style code w_0 and a small offset vector o_i for each of StyleGAN’s 18 layers. The final style for layer i is $w_0 + o_i$. During training, two principles are enforced:

1. Progressive training: initially predict only w_0 then gradually allow one layer’s offset at a time, progressing from coarse to fine layers.
2. Offset regularization: penalize the magnitude of the offsets (via L2) to keep $w_0 + o_i$ close to w_0 and thus close to the learned W manifold. A small latent discriminator (GAN loss in latent space) is also trained to distinguish real W samples from the encoder’s outputs.

During training e4e uses the same image-space losses as pSp (L2, LPIPS, identity) and adds the above editability losses. The result is an encoder that strikes a balance: it produces

reconstructions nearly as good as pSp, but the codes are “closer to W ” so that semantic edits transfer more reliably. e4e still outputs an extended style code, but with built-in bias towards W . In practice one can edit e4e codes using the same methods (adding latent directions, style mixing), and edits tend to be more robust. For faces, e4e also incorporates a cosine identity loss (ArcFace or general ResNet) to preserve identity under reconstruction.

e4e greatly improves editability. It was shown to maintain coherent attribute manipulations (age, expression, pose) on real images better than pSp, with only a minor increase in reconstruction error. Its inference speed is still one forward pass (plus negligible overhead from the latent discriminator at training time). It generalizes to multiple domains (faces, cars, etc.) by using a domain-agnostic feature extractor for identity.

To achieve editability, e4e accepts a small drop in reconstruction fidelity. Very fine details may be slightly lost compared to pSp. Also, the progressive training and latent GAN add complexity, and tuning the trade-off weights requires care. The notion of “proximity to W ” is somewhat loosely defined, so extremely out-of-distribution images (e.g. non-human faces) may still not be perfectly encoded. Tov et al. (2021)

2.5.3 ReStyle

ReStyle (ICCV 2021) focuses on improving inversion accuracy by iterative refinement. Instead of a single encoder pass, ReStyle’s encoder is applied multiple times in a feedback loop. Starting from an initial latent (the average w vector), ReStyle repeatedly concatenates the original image with the generator’s current reconstruction and feeds them into the encoder. At each iteration t , the encoder predicts a residual $\Delta w^{(t)}$ to add to the current latent.

$$\text{At step } t, w^{(t)} = w^{(t-1)} + E(x, G(w^{(t-1)}))$$

,

with $w^{(0)}$, set to the average latent, and $G(w)$, the current output image. The encoder is trained to minimize reconstruction loss at each step (summing L2+LPIPS over all iterations).

ReStyle can be applied on top of any base encoder (e.g. pSp or e4e). In implementation, the authors actually simplify those encoders by using only the final feature map (no pyramid). Each iteration uses a fresh forward pass through the same network (weights are shared across steps). The training, therefore, unrolls a small multi-step network with losses on each reconstructed image.

ReStyle yields a more accurate latent code after its iterations, meaning $G(w)$ more closely matches the input. This generally improves any downstream editing (since the base reconstruction is better). In other words, it aims to match the quality of per-image optimization (but much faster). The final code is in W^+ and can be edited normally.

ReStyle significantly improves reconstruction quality compared to standard feed-forward encoders. It achieves near-optimization accuracy with only a small inference-time cost (typically 3-5x forward passes). Because it uses the same encoder architecture at each step, it avoids complex modifications. It can even be viewed as a learned optimization in latent space.

The iterative loop means inference is slower (several passes). Although small (e.g. 5 steps), it is no longer a single-shot encoder. Also, ReStyle does not explicitly enforce editability constraints - it focuses on fidelity. It inherits the latent space choice of the base encoder: if initialized in W^+ the result can still lie outside the original W distribution. Finally, training is more complex (unrolling multiple steps with deep supervision) and may be sensitive to number of iterations. (Alaluf et al., 2021)

2.5.4 StyleFeatureEditor

StyleFeatureEditor (SFE) introduces a new latent representation combining StyleGAN’s usual style codes, w or w_i , with a feature tensor F_k from an intermediate generator layer. Intuitively, the feature tensor F_k , high-dimensional spatial features, captures fine detail that W^+ alone may miss. The method has two networks: an inverter I and a feature editor H . The inverter I takes an image X and predicts both a style code $w \in W^+$ and a feature map F_k . Specifically, I first maps X to w via linear layers, and also to an intermediate feature F_{pred} . Then it fuses

F_{pred} with the actual generator’s feature. to produce the final inversion feature F_k .

Crucially, StyleFeatureEditor trains a separate editor network $H(F, \Delta)$ that adjusts the feature tensor for editing. Given an editing direction d in n latent space, one computes the change $\Delta = F_k(w + d) - F_k(w)$ by sampling from a pretrained encoder. Then H is trained to take the inverted feature F_k and Δ and output the edited feature $F'_k = H(F_k, \Delta)$. In inference, for a desired edit d , one computes $w' = w + d$, uses H to updated $F \Rightarrow F'_k$ and generates the edited image from w', F'_k .

StyleFeatureEditor undergoes 2 phase training:

1. Inverter I is trained on real images with standard losses (L2+LPIPS+identity, plus latent regularization and adversarial loss).
2. Editor H is trained on synthetic pairs: for each real X , a pretrained W^+ encoder produces w , an edit d is sampled, yielding, X'_E by adjusting H (inverter frozen).

StyleFeatureEditor achieves extremely high reconstruction quality and preserves detail even under editing. In experiments it produces nearly indistinguishable inverted images, and crucially it “preserves most” of those fine details when editing. By operating directly on feature maps, it can maintain textures and fine structure that pure latent edits often lose. It is capable of editing challenging out-of-domain examples and does not rely solely on W or W^+ for content. (Bobkov et al., 2024)

2.6 Few-Shot Image Generation in GANs

Few-shot image generation (FSIG) extends few-shot learning to generative modeling: the goal is to synthesize novel images for a new category given only a handful of training examples. This contrasts with few-shot classification, where techniques like transfer learning have proven effective (Chowdhury et al., 2021). In FSIG, however, naively fine-tuning a powerful generative model on few examples leads to overfitting and mode collapse. According to Zhao et al. (2023), modern GANs “tend to replicate the training samples” when adapted to only 10 target images.

As a result, FSIG methods typically transfer rich prior knowledge from a large source domain and then adapt carefully to the new domain. This is often framed as a style-transfer or domain-adaptation problem: a generator pretrained on a large dataset ,like FFHQ faces, is fine-tuned on a small target set without access to the source data (Cao and Gong, 2024; Zhao et al., 2023). The key challenge is preserving the source model’s diversity and high-fidelity output while learning the new style from very limited data.

Generative Adversarial Networks (GANs) have been the dominant framework for FSIG. The standard paradigm is to take a GAN (often StyleGAN or BigGAN) pretrained on a large source dataset and adapt it to the new class. Early approaches simply fine-tuned all weights on the few-shot target (TGAN) (Zhao et al., 2023), but this quickly overfits (the generator memorizes the few examples). Numerous strategies have been proposed to prevent collapse and preserve diversity during adaptation. or instance, FreezeD freezes some high-resolution layers of the discriminator, ADA (adaptive data augmentation) introduces random augmentations during training, and EWC applies an Elastic Weight Consolidation penalty on important weights (Mo et al., 2020). Another approach Cross-domain Correspondence (CDC) preserves the relative distances between different source-domain samples in the target embedding space (Ojha et al., 2021). In practice, Zhao et al. (2022) found that most of these GAN-adaptation methods converge to similar image quality after many iterations – the real difference is how slowly they lose diversity during training (Zhao et al., 2023). In other words, diversity degradation (mode collapse) is the main bottleneck: methods that slow down this collapse (e.g. by freezing parameters or using strong regularization) tend to perform better overall.

Building on this insight, Zhao et al. propose a contrastive learning scheme to retain diversity. Their Dual Contrastive Learning (DCL) maximizes mutual information between the source and target generators: the idea is that the same latent input z should map to semantically corresponding images before and after adaptation. Concretely, they use a contrastive loss on multi-level features of the generator and discriminator to enforce that source and target outputs remain similar in high-level content. This preserves the source generator’s rich

variation in the adapted model, yielding state-of-the-art results on several datasets (Zhao et al., 2023). Similarly, Hong et al. (2020) introduce DeltaGAN, which decomposes generation into a reconstruction sub-network (that computes an intra-class “delta” transformation between two samples of the same category) and a generation sub-network (that applies a sample-specific delta to a base image) (Hong et al., 2022). DeltaGAN’s adversarial “delta matching” encourages high diversity by learning many distinct transformations from the limited examples.

GAN-based FSIG methods share a transfer-learning theme: leverage a pretrained model’s knowledge and carefully regularize the adaptation. Typical training strategies include fine-tuning, parameter freezing, data augmentation, and contrastive or distance-preserving losses. Their success is usually measured by metrics like Fréchet Inception Distance (FID) or LPIPS (perceptual diversity).

Transfer learning is a commonly used training strategy in generative models that starts from a pre-trained model (GAN, VAE, or diffusion) and updates it using a few target images. Pure fine-tuning can often overfit the model, making it difficult to generalize well (Zhao et al., 2023). Meta-learning involves training on simulated few-shot tasks so that it can adapt quickly. Contrastive/Distance Losses are used to enforce that the adapted generator preserves relationships between different domains. Data augmentation and synthesis create pseudo-examples by replacing source images with those in the target fine-tuning process. While some GANs/VAEs may use MAML or Reptile, most often they still need per-task fine-tuning and struggle to generate sharp images (Hong et al., 2022). In contrast, meta-learned GANs/VAEs have been attempted but are usually limited by their overfitting problem (Hong et al., 2022). Meta-Learning involves training on many simulated few-shot tasks so that it can adapt quickly. Data augmentation and synthesis create pseudo-examples by replacing source images with those in the target fine-tuning process. In this way, humans can better understand how to generate realistic images based on the data they have available.

Evaluation strategies are also noteworthy. FSIG is evaluated on cross-domain tasks such as

generating artistic or cartoon versions of FFHQ faces, or adapting LSUN Church to “Haunted House” style, or similar sketch-to-photo tasks. Metrics include FID for quality (when enough data exists) and LPIPS or recall for diversity. Zhao et al. (2023) emphasize that with only 10 real images, FID is unstable, so they instead use intra-LPIPS (average pairwise LPIPS of generated set) to quantify diversity .

Despite these advances, few-shot image generation remains challenging. Surveys note that learning from very few or zero examples is still “a serious challenge” (Song et al., 2022). Key open issues include: catastrophic forgetting (the model loses the source distribution entirely), mode collapse (outputs lack diversity), and evaluation (how to measure quality/diversity with tiny real sample sets). Overfitting to the few examples is ubiquitous: even with strong regularization, generators tend to memorize salient features. Furthermore, different generative families have trade-offs – GANs are fast and high-quality but brittle, VAEs are stable but blurrier, and diffusion models are powerful but computationally heavy. A brief comparison: GANs excel at sharp images but require careful balancing of losses; VAEs naturally encourage covering the whole distribution (which aids diversity) at the cost of some sharpness; diffusion models sample iteratively and often capture fine detail, but naively adapting them can be even more parameter-sensitive than GANs (Cao and Gong, 2024).

CHAPTER 3 METHODOLOGY

3.1 Research Design

This study, “FeatureSAGE: Stable Attribute Group Editing With Integrated StyleFeatureEditing Encoder for Enhanced Few-Shot Image Generation,” adopts a quantitative experimental research design to evaluate improvements to the existing SAGE framework. In particular, the GAN inversion module of SAGE is modified by integrating the StyleFeatureEditor (SFE) inverter and its Feature Editor module. The modification is evaluated under a one-shot setting, where attribute manipulation is conditioned on a single reference image per test instance. Objective evaluation is performed using Fréchet Inception Distance (FID) and Learned Perceptual Image Patch Similarity (LPIPS), which measure distributional image quality and perceptual variation, respectively.

3.1.1 Experimental Setup

The experimental setup is designed to evaluate the effectiveness of the proposed FeatureSAGE framework in comparison to the existing SAGE method. The evaluation follows a quantitative experimental design that involves controlled changes to the inversion and feature-editing components and the measurement of model outputs using standardized metrics.

- **Independent Variable:** The main experimental manipulation is the GAN inversion method. The baseline SAGE configuration uses the original pixel2style2pixel (pSp) encoder, whereas the proposed FeatureSAGE configuration integrates the StyleFeatureEditor inverter and Feature Editor module.
- **Dependent Variables:** To objectively evaluate the effectiveness of the GAN inversion and editing pipeline, the following metrics are employed:

- Reconstruction quality and generated-image distribution alignment, measured using Fréchet Inception Distance (FID).
- Perceptual similarity or perceptual diversity, depending on the evaluation context, measured using Learned Perceptual Image Patch Similarity (LPIPS).

- **Controlled Variables**

1. **Generator Architecture:** A fixed, pretrained StyleGAN2 generator is used in both configurations for consistent image synthesis capability.
2. **Training Dataset:** Both configurations use the same FUNIT Animal Faces and 102 Flowers datasets with the same preprocessing and seen/unseen class splits.
3. **Evaluation Protocol:** The same unseen categories, one-shot support samples, query sets, and evaluation procedures are used for both configurations.
4. **Loss Functions:** Reconstruction and editing losses are applied consistently within the corresponding training phases.
5. **Optimization Settings:** Both configurations use the Adam optimizer and the fixed learning-rate settings specified in the training strategy.
6. **Evaluation Metrics:** FID and LPIPS are computed using the same implementations and parameters.
7. **Hardware and Software Environment:** All experiments are conducted using the same GPU-enabled PyTorch environment to avoid performance inconsistencies.

- **Experimental Conditions**

Two experimental configurations are used:

- SAGE (Baseline): Employs the original pSp encoder with StyleGAN2.
- FeatureSAGE (Proposed): Integrates the StyleFeatureEditor inverter and Feature Editor.

3.1.2 Hardware and System Requirements

All experiments were conducted on a virtual machine instance with GPU pass-through hosted on institutional cloud infrastructure. The same hardware and software configuration was used for the baseline and proposed models so that performance differences could be attributed to the inversion and feature-editing components rather than to changes in the execution environment. Table 3.1 summarizes the system requirements reported for the implementation.

Component	Specification
CPU	QEMU Virtual CPU version 2.5+, 2 processors at 2.19 GHz
RAM	32.0 GB
GPU	NVIDIA GRID V100D-32A, virtualized Tesla V100 with 16 GB HBM2
Storage	Not specified in the source implementation notes; storage is required for the datasets, model checkpoints, and generated outputs.
Software and frameworks	PyTorch environment; pretrained StyleGAN2 generator; pSp encoder for the baseline and editing-pair generation; StyleFeatureEditor inverter and Feature Editor; MTCNN for animal-face cropping and alignment; Adam optimizer; mixed precision for Feature Editor training; FID and LPIPS metric implementations.

Table 3.1 Hardware and system requirements used in the experimental setup.

The GPU specification is important because FeatureSAGE performs multiple generator passes and feature-tensor operations during inversion and editing. The use of mixed precision in the Feature Editor phase reduces memory pressure on the 16 GB V100 vGPU while preserving the same model configuration and reported evaluation protocol.

3.1.3 Few-Shot Experimental Setting

In this study, the term *few-shot* refers to the number of reference images available per target class used to construct class-level latent representations for attribute group editing and downstream evaluation. A **one-shot** setting utilizes a single reference image per target class, representing an extreme data-scarce scenario in which attribute editing and inversion must be performed with minimal visual information. This one-shot configuration is applied consistently across all datasets and experimental conditions for both the baseline SAGE and the proposed FeatureSAGE framework. By explicitly controlling the evaluation to a single reference image per class, the experimental protocol enables a systematic assessment of FeatureSAGE’s robustness, stability, and generalization capability under severe data limitations. The few-shot constraint is applied during test-time adaptation and evaluation on unseen classes, while the StyleGAN2 generator and attribute directions are learned from seen-category data.

3.2 Data Preprocessing

3.2.1 Data Sources

This research uses publicly available datasets containing structured visual categories for few-shot image generation and attribute editing. The datasets are selected because they contain clearly grouped classes and visual variation that can be used to evaluate whether generated images preserve category-level structure while allowing appearance changes. Only a limited number of representative samples per class are used during few-shot evaluation to reproduce the data-scarce setting of the study. The datasets used are:

1. FUNIT Animal Faces Dataset

Figure 3.1 shows representative samples from the FUNIT Animal Faces dataset. The dataset is composed of 117,484 animal face images at 512×512 resolution and is organized into 149 classes, where each class corresponds to one animal species. Each



Figure 3.1 Sample images from the FUNIT Animal Faces dataset. Source: (Liu et al., 2019a)

domain includes a diverse range of species and breeds, ensuring variation in appearance while maintaining consistent facial alignment to support generative model training. The images were curated from openly available sources and standardized in resolution and alignment, with the eyes centrally positioned to provide consistency for learning. A subset of the images is reserved as a test set, with the remainder used for training. The diversity and quality of the FUNIT Animal Faces dataset make it a strong benchmark for generative and attribute-editing models in few-shot settings (Liu et al., 2019a).

2. 102 Flowers Dataset



Figure 3.2 Sample images from the 102 Flowers dataset. Source: (Nilsback and Zisserman, 2008)

Figure 3.2 shows sample images from the 102 Flowers dataset, which comprises 102 flower classes found in the United Kingdom. Each class contains between 40 and 258 images. The dataset includes variation in flower size, pose, and lighting conditions, and it contains both large intra-class differences and visually similar categories. These characteristics make the dataset suitable for evaluating fine-grained category preservation and appearance variation. The raw images have varying dimensions; for example, one raw image from the dataset measures 500×667 pixels (Nilsback and Zisserman, 2008).

3.2.2 Data Preprocessing Steps

The same preprocessing pipeline is applied across the baseline SAGE and proposed FeatureSAGE configurations:

- **Resizing and normalization:** Images are resized according to the StyleGAN2 configuration used in the experiment and are normalized to the range $[-1, 1]$ before inversion or synthesis. The quantitative results reported in Chapter 4 are evaluated at 1024×1024 resolution.
- **Filtering:** Extremely low-quality or mislabeled images are removed before training and evaluation. For the FUNIT Animal Faces dataset, MTCNN is used to crop and horizontally align animal faces so that the eyes are leveled.
- **Train-test splitting:** The datasets are partitioned into seen categories for model fitting and unseen categories for few-shot testing:
 - **Animal Faces:** Out of the total 149 categories, 119 are used as seen classes for training, while the remaining 30 serve as unseen classes for evaluation.
 - **Flowers:** Out of the total 102 categories, 85 are assigned to the seen training split and 17 are reserved for the unseen test split.
- **Augmentation:** No augmentation is used beyond centering and flipping, since the study focuses on evaluating the same domains under controlled few-shot conditions.

3.3 Model Architecture

This study proposes a modified SAGE framework named **FeatureSAGE**. The architecture combines a pretrained StyleGAN2 generator, the StyleFeatureEditor inverter and Feature Editor, and SAGE’s attribute group editing mechanism. The purpose of the modification is to improve the inversion stage of SAGE by representing each input image with both a layer-wise latent

code and an intermediate feature tensor. This design follows the observation of Ding et al. (2023) that stronger inversion can further improve the potential of SAGE for generative augmentation, and it follows Bobkov et al. (2024) in using both W^+ latents and feature latents for detail-preserving editing.

The main components of FeatureSAGE are:

1. **StyleGAN2 generator:** The pretrained StyleGAN2 generator synthesizes images from latent codes. In FeatureSAGE, the generator receives the edited W^+ latent code together with an edited intermediate feature tensor produced by the Feature Editor.
2. **StyleFeatureEditor inverter:** The inverter replaces the original pSp inversion module used in SAGE. For each input image, it predicts a W^+ latent code and an intermediate feature tensor. The W^+ code captures layer-wise style information, while the feature tensor carries spatial details that are difficult to preserve using latent codes alone.
3. **Feature Editor:** The Feature Editor modifies the extracted feature tensor according to the latent-space edit. This allows the decoder path of the generator to synthesize images whose semantic edit is controlled by the latent code while finer details are supported by feature-level information.
4. **SAGE attribute group editing:** The attribute factorization, class embedding mapping, and adaptive editing blocks are retained from SAGE. These blocks estimate category-relevant embeddings and category-irrelevant directions so that unseen classes can be edited under a one-shot setting.

3.3.1 Framework Overview

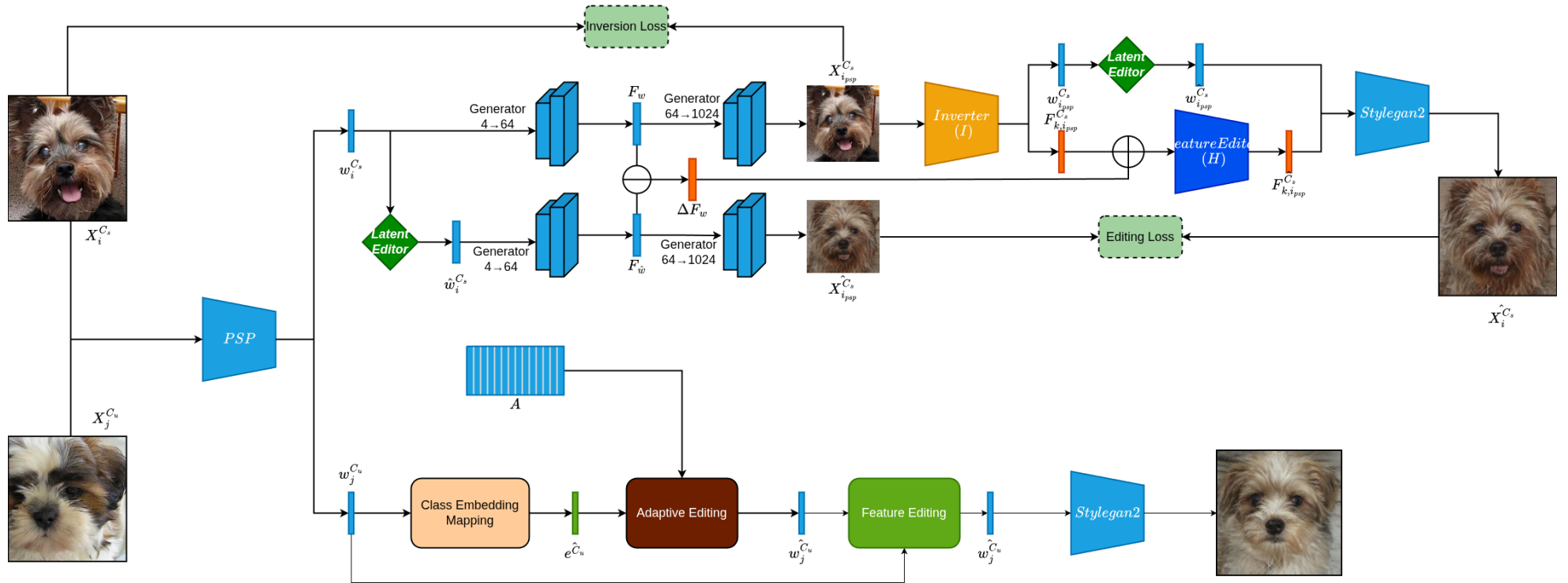


Figure 3.3 FeatureSAGE framework

Figure 3.3 shows the overall FeatureSAGE framework. The training set for seen categories and the testing set for unseen categories are represented in Equation 1:

$$D_{\text{train}} = \{x_i^{C_s}\}_{i=1}^{N_s}, \quad D_{\text{test}} = \{x_i^{C_u}\}_{i=1}^{N_u}, \quad (1)$$

where $x_i^{C_s}$ denotes an image from a seen category C_s , $x_i^{C_u}$ denotes an image from an unseen category C_u , and N_s and N_u denote the number of samples used from the seen and unseen splits, respectively. Equation 1 provides the notation used throughout the methodology for separating classes used in training from classes used during one-shot evaluation.

3.3.2 Feature Extraction and Latent-Space Representation

Feature extraction occurs before the generator synthesizes an output image. In the baseline SAGE configuration, the pSp encoder maps an input image into the extended StyleGAN2 latent space, as shown in Equation 2:

$$PSP(x_i^{C_s}) = w_i^{C_s}, \quad w_i^{C_s} \in W^+. \quad (2)$$

In Equation 2, $PSP(\cdot)$ is the pSp encoder and $w_i^{C_s}$ is the predicted W^+ latent code for a seen-category image. The W^+ space is important because it contains separate style vectors for the generator layers, allowing the model to represent coarse-to-fine visual attributes.

In FeatureSAGE, the StyleFeatureEditor inverter extracts two representations from an image, as defined in Equation 3:

$$I(x_i^C) = (w_i^C, F_{k,i}^C), \quad w_i^C \in W^+. \quad (3)$$

Here, $I(\cdot)$ is the StyleFeatureEditor inverter, w_i^C is the layer-wise latent code, and $F_{k,i}^C$ is the intermediate feature tensor extracted at generator layer k . The document uses the ninth StyleGAN feature layer for feature injection. The latent code represents semantic and style-level information, while the feature tensor represents spatial detail and texture

information. These two extracted features are significant because the latent code controls the edit direction, whereas the feature tensor helps preserve local details during synthesis.

The generator can also expose intermediate features for an original and edited latent code. Equation 4 defines these generator-side features:

$$(x_{i,\text{psp}}^{C_s}, F_w) = G(w_i^{C_s}), \quad (\hat{x}_{i,\text{psp}}^{C_s}, F_{\hat{w}}) = G(\hat{w}_i^{C_s}). \quad (4)$$

In Equation 4, $G(\cdot)$ is the StyleGAN2 generator, F_w is the intermediate feature tensor associated with the original latent code, and $F_{\hat{w}}$ is the feature tensor associated with the edited latent code. These feature tensors are later used to compute the feature-space change required by the Feature Editor.

The latent space in FeatureSAGE therefore contains three related types of information: the W^+ latent code w , the category-level class embedding e^C , and the intermediate feature tensor F_k . The W^+ code encodes layer-wise style information; the class embedding stores category-relevant information through an average latent representation; and F_k stores detail-rich spatial information used by the decoder block. Their combined use addresses the fidelity-editability trade-off identified in the literature without changing the reported experimental claims.

3.3.3 Attribute Factorization

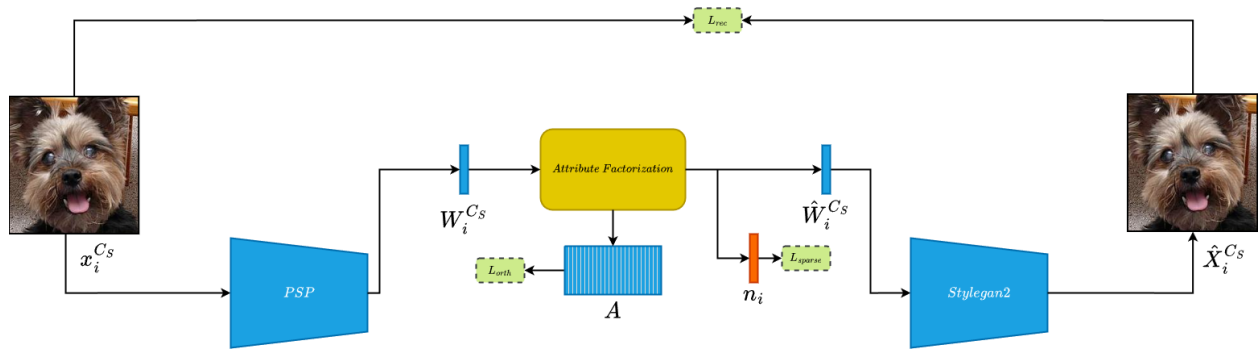


Figure 3.4 Attribute Factorization Training Framework (Ding et al., 2023)

To enable disentangled editing, FeatureSAGE retains the SAGE attribute factorization

block. The block learns category-irrelevant directions by comparing the latent code of a sample with the mean latent code of its category. The class embedding for a category is computed as the mean of its latent codes, as shown in Equation 5:

$$e^C = \frac{1}{N_C} \sum_{i=1}^{N_C} w_i^C, \quad (5)$$

where e^C is the class embedding for category C , N_C is the number of latent codes available for the category, and w_i^C is the latent code of the i th image. Equation 5 provides the category-relevant center used to separate identity or class-level information from editable appearance variation.

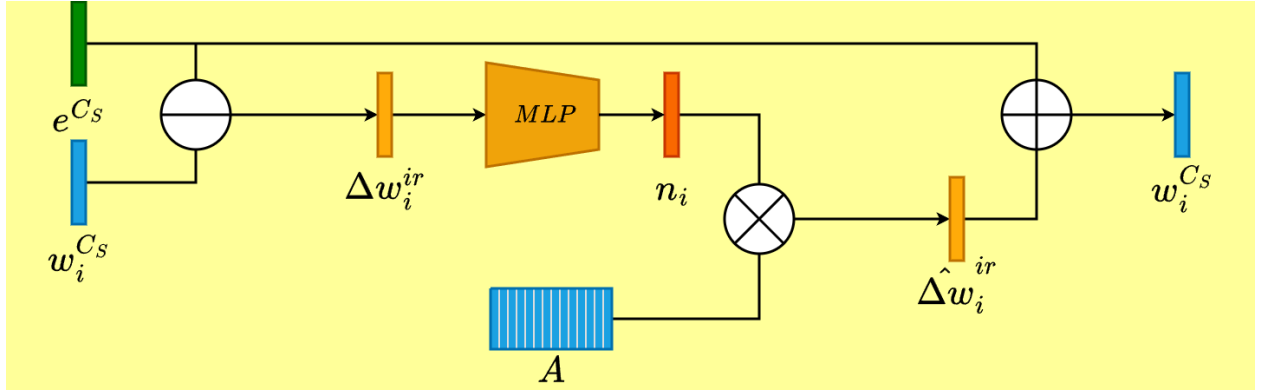


Figure 3.5 Attribute Factorization Block, derived from (Ding et al., 2023)

In the Attribute Factorization block shown in Figure 3.5, the category-irrelevant residual is computed in Equation 6:

$$\Delta w_i^{ir} = w_i^{Cs} - e^{Cs}. \quad (6)$$

In Equation 6, Δw_i^{ir} is the latent residual that remains after subtracting the class embedding e^{Cs} from the sample latent code w_i^{Cs} . The purpose of this residual is to isolate appearance attributes that affect how an object looks without changing its class membership, such as orientation, color, background, texture, and expression.

The residual is then reconstructed using a learned sparse dictionary, as shown in Equation 7:

$$\hat{\Delta w}_i^{ir} = An_i. \quad (7)$$

In Equation 7, A is the global dictionary of category-irrelevant directions and n_i is the sparse coefficient vector predicted for the image. The dictionary basis follows sparse dictionary learning: each image is represented using only a small number of active directions so that individual directions remain more semantically interpretable.

Category-irrelevant editing is defined in Equation 8. The edit changes appearance while preserving the category:

$$edit_{ir}(G(W_i^{C_s})) = G(W_i^{C_s} + \alpha \Delta W^{ir}, H(F_{k,i}, \alpha \Delta W^{ir})) = \hat{X}_i^{C_s}. \quad (8)$$

In Equation 8, $edit_{ir}(\cdot)$ denotes category-irrelevant editing, α controls the edit intensity, ΔW^{ir} is the category-irrelevant direction, $H(\cdot)$ is the Feature Editor, and $\hat{X}_i^{C_s}$ is the edited image. The theoretical basis is the SAGE assumption that category-irrelevant directions can be learned from seen categories and transferred to unseen categories.

The sparse dictionary learning objective is stated in Equation 9:

$$\min_{n_i} \|n_i\|_0 \quad \text{s.t.} \quad \Delta W_i^{ir} = An_i. \quad (9)$$

In Equation 9, $\|n_i\|_0$ counts the number of nonzero entries in n_i . The purpose of the L_0 constraint is to encourage sparse use of the dictionary so that each generated edit is controlled by a limited number of directions.

In practice, SAGE optimizes this factorization with an encoder-decoder structure in which a Multi-Layer Perceptron (MLP) predicts the sparse representation, as shown in Equation 10:

$$n_i = MLP(\Delta W_i^{ir}). \quad (10)$$

Equation 10 defines how the residual latent vector is encoded into the sparse coefficient

vector used by the dictionary A .

3.3.4 Class Embedding Mapping

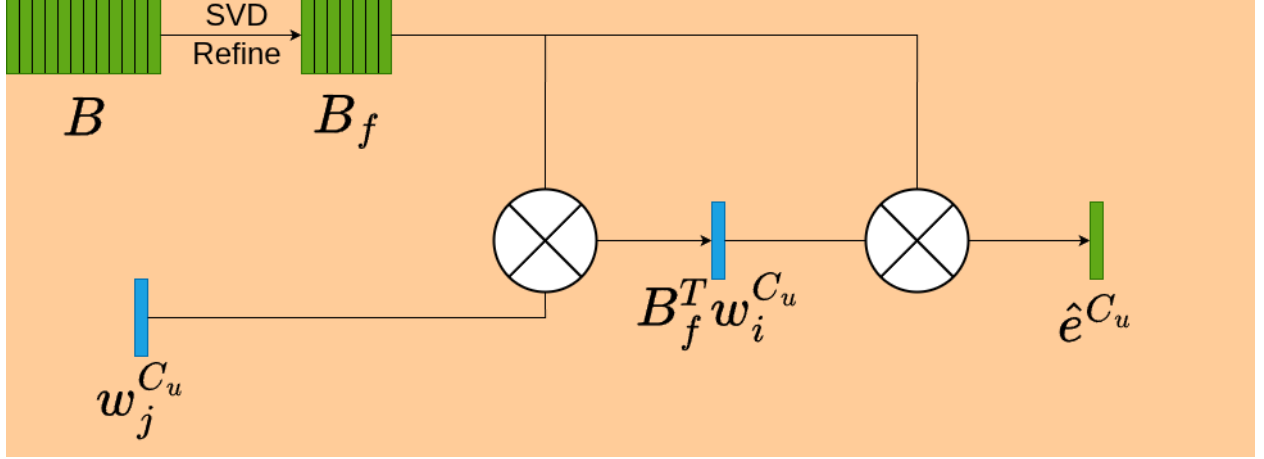


Figure 3.6 Class Embedding Block, adapted from (Ding et al., 2023)

Compared with Attribute Group Editing (AGE) (Ding et al., 2022), which randomly samples editing directions from A , SAGE uses class embedding mapping. The class embedding mapping block shown in Figure 3.6 identifies the most important directions in the latent-code space for each category group and uses those directions to guide subsequent edits. This part of the framework remains unchanged in FeatureSAGE.

The class embedding set for the seen categories is denoted as B . Since the embeddings are computed from average latent codes, B may still contain interference from category-irrelevant attributes. To filter the category-relevant subspace, singular value decomposition (SVD) is applied as shown in Equation 11:

$$B = U_B \Sigma_B V_B^T. \quad (11)$$

In Equation 11, B is the matrix of category-relevant class embeddings, U_B contains the left singular vectors, Σ_B contains singular values in descending order, and V_B^T is the transpose of the right singular-vector matrix. The top- t_B directions in U_B are selected as the final category-relevant attribute dictionary $B_f \in \mathbb{R}^{18 \times 512 \times t_B}$.

The class embedding of an unseen category is estimated using the filtered dictionary, as shown in Equation 12:

$$\hat{e}^{C_u} = \frac{1}{N_u} \sum_{i=1}^{N_u} B_f B_f^T W_i^{C_u}. \quad (12)$$

In Equation 12, $\hat{e}^{C_u}$ is the estimated embedding for unseen category C_u , N_u is the number of latent codes available for that unseen category, and $W_i^{C_u}$ is the latent code of the i th unseen-category sample. The projection $B_f B_f^T W_i^{C_u}$ retains category-relevant components while suppressing most category-irrelevant variation.

3.3.5 Adaptive Editing

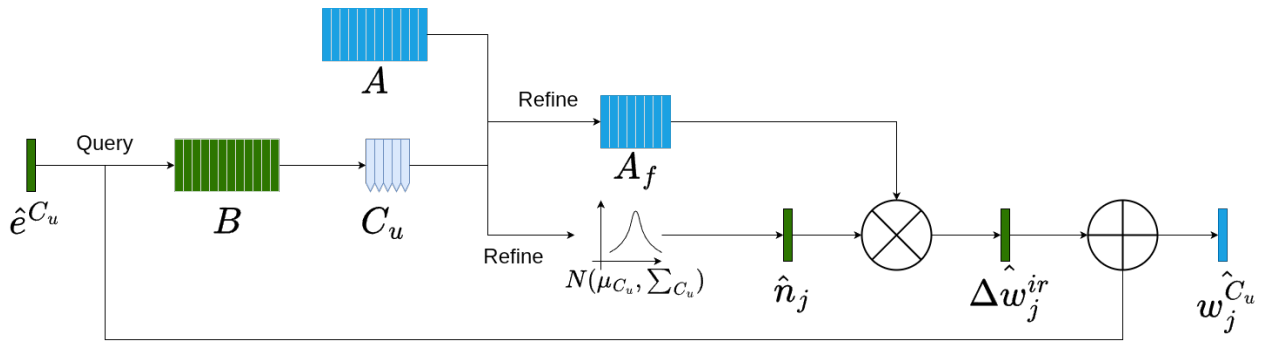


Figure 3.7 Adaptive Editing Block, from (Ding et al., 2023)

Figure 3.7 shows the adaptive editing block of FeatureSAGE. Category-irrelevant qualities typically have similar distributions in closely related categories but may differ across distant categories. Therefore, adaptive editing selects seen categories that are close to an unseen category in the class-embedding space and uses their common editing directions to guide generation.

To identify the category-irrelevant editing directions that best fit an unseen category C_u , the residual direction is back-projected into the sparse representation, as shown in Equation 13:

$$\hat{n}_i = A^\dagger \Delta W_i^{ir}. \quad (13)$$

In Equation 13, $A^\dagger$ denotes the pseudo-inverse of the dictionary A , ΔW_i^{ir} is the category-irrelevant residual, and $\hat{n}_i$ is the estimated sparse representation. The pseudo-inverse is used to express an observed residual in the coordinate system of the learned dictionary.

The mean absolute activation of sparse directions across the top- t_C similar seen categories is computed in Equation 14:

$$\overline{|n|}^{C_u} = \frac{1}{t_C} \sum_{t=1}^{t_C} \frac{1}{N_{S_t}} \sum_{i=1}^{N_{S_t}} |\hat{n}_i^{C_{S_t}}|. \quad (14)$$

In Equation 14, t_C is the number of similar seen categories, N_{S_t} is the number of samples in the t th selected seen category, and $\hat{n}_i^{C_{S_t}}$ is the sparse representation of the i th sample from that category. The result $\overline{|n|}^{C_u}$ measures which dictionary directions are frequently or strongly activated for categories similar to C_u . For each layer in W^+ , the top- t_A directions are selected from A to form the adaptive category-irrelevant dictionary $A_{C_u} \in \mathbb{R}^{18 \times 512 \times t_A}$.

SAGE assumes that the sparse representations for related categories follow a Gaussian distribution. This distribution is estimated in Equation 15:

$$\tilde{n}_j \sim \mathcal{N}(\mu_{C_u}, \Sigma_{C_u}). \quad (15)$$

In Equation 15, μ_{C_u} and Σ_{C_u} are the estimated mean and covariance of sparse codes for the selected categories, and $\tilde{n}_j$ is a sampled sparse code used to generate a new image. This sampling step provides controlled diversity for the unseen category.

The final adaptive generation equation is given in Equation 16:

$$X_j^{C_u} = G(\hat{e}^{C_u} + \alpha A_{C_u} \tilde{n}_j, H(F_{k,j}^{C_u}, \alpha A_{C_u} \tilde{n}_j)). \quad (16)$$

In Equation 16, $X_j^{C_u}$ is a generated image for unseen category C_u , $\hat{e}^{C_u}$ is the estimated class embedding, A_{C_u} is the adaptive dictionary, $\tilde{n}_j$ is the sampled sparse code, α controls the strength of the edit, and $H(\cdot)$ edits the feature tensor used by the generator. This equation summarizes how FeatureSAGE transforms one-shot class information and sampled attribute

variation into a generated image.

3.3.6 Feature Editing Decoder Block

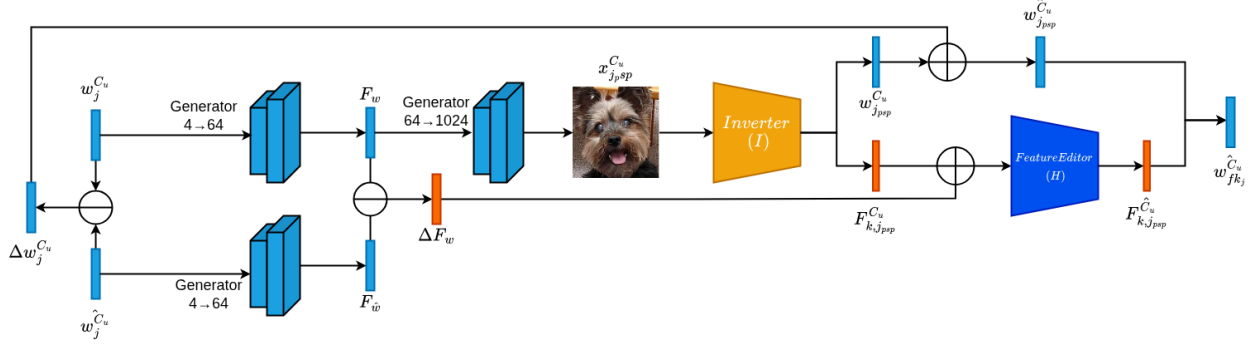


Figure 3.8 Feature Editing Decoder Block

Figure 3.8 shows the Feature Editing Decoder Block. The block receives the original latent representation, the edited latent representation produced by adaptive editing, and the intermediate feature tensor extracted by the StyleFeatureEditor inverter. Its role is to transform the input features into an edited feature tensor that can be injected into the ninth StyleGAN feature layer during image synthesis.

The latent-space difference used to describe the semantic edit is defined in Equation 17:

$$\Delta w_j^{C_u} = \hat{w}_j^{C_u} - w_j^{C_u}. \quad (17)$$

In Equation 17, $w_j^{C_u}$ is the original latent code and $\hat{w}_j^{C_u}$ is the edited latent code. This difference specifies the semantic movement applied in W^+ .

The feature-space difference used for feature injection is defined in Equation 18:

$$\Delta F_w = F_w - F_{\hat{w}}. \quad (18)$$

In Equation 18, F_w is the intermediate feature tensor produced from the original latent code and $F_{\hat{w}}$ is the corresponding tensor produced from the edited latent code. The decoder block uses this difference to determine how the original feature tensor should be modified to

match the target semantic edit.

The Feature Editor transformation is represented in Equation 19:

$$F_{k,j}^{\text{edit}} = H(F_{k,j}^{C_u}, \Delta F_w). \quad (19)$$

In Equation 19, $F_{k,j}^{C_u}$ is the feature tensor extracted by the inverter for the unseen-category input, ΔF_w is the feature difference from Equation 18, and $F_{k,j}^{\text{edit}}$ is the edited tensor injected into the generator. The decoder therefore modifies the feature-level representation while the edited latent code controls the global semantic direction. The output image is then synthesized by the generator using the edited latent code and the edited feature tensor, as shown in Equation 20:

$$\hat{X}_j^{C_u} = G(\hat{w}_j^{C_u}, F_{k,j}^{\text{edit}}). \quad (20)$$

Equation 20 completes the decoder pathway: input image features and latent edits are transformed into an edited high-resolution output.

3.3.7 Training Strategy

In the training stage, a StyleGAN2 generator is trained using images from the seen categories. The StyleFeatureEditor inverter and the pSp encoder are also trained or used according to their roles in the pipeline. To manage resource consumption, FeatureSAGE uses a two-phase strategy: attribute factorization training followed by Feature Editor training.

3.3.8 Phase 1: Attribute Factorization Training

Phase 1 trains the attribute factorization block for disentangled sampling of category-irrelevant directions. The following losses are applied to the attribute factorization block.

- **Sparsity Loss** (L_{sparse}): The sparsity loss encourages the learned attribute code to

activate only a small number of dictionary directions. Equation 21 defines the loss:

$$L_{\text{sparse}} = \|\sigma(\theta_0 n_i - \theta_1)\|_1. \quad (21)$$

In Equation 21, $\sigma(\cdot)$ denotes the sigmoid function, n_i is the sparse code, and θ_0 and θ_1 are hyperparameters controlling sparsity. The sigmoid compresses the MLP output and the L_1 norm approximates sparse behavior in a differentiable way.

- **Reconstruction Loss (L_{rec}):** The reconstruction loss anchors the edited output to the input image so that edits do not destroy the original content. Equation 22 defines this loss:

$$L_{\text{rec}} = \|G(e^{C_s} + An_i) - x_i^{C_s}\|_2. \quad (22)$$

In Equation 22, e^{C_s} is the class-center latent, An_i is the attribute offset, $G(\cdot)$ is the generator, and $x_i^{C_s}$ is the input image. The loss penalizes large differences between the generated reconstruction and the original image.

- **Orthogonality Loss (L_{orth}):** The orthogonality loss discourages category-irrelevant directions from overlapping with category-relevant directions. Equation 23 defines this loss:

$$L_{\text{orth}} = \|B^T A\|_F^2. \quad (23)$$

In Equation 23, A is the category-irrelevant dictionary, B is the category-relevant embedding basis, and $\|\cdot\|_F^2$ denotes the squared Frobenius norm. This term preserves class identity by reducing leakage between the category-relevant and category-irrelevant subspaces.

The overall attribute factorization loss is defined in Equation 24:

$$L_{\text{af}} = L_{\text{rec}} + \lambda_1 L_{\text{orth}} + \lambda_2 L_{\text{sparse}}. \quad (24)$$

In Equation 24, λ_1 and λ_2 are weights for the orthogonality and sparsity losses (Ding et al., 2023). Training uses the Adam optimizer with a fixed learning rate of 1×10^{-4} , $\lambda_1 = 5 \times 10^{-4}$, $\lambda_2 = 5 \times 10^{-3}$, $\theta_0 = 5 \times 10^{-1}$, and $\theta_1 = -1$.

3.3.9 Phase 2: Feature Editor Training

Phase 2 trains the Feature Editor H using editing pairs generated with a pretrained pSp encoder. Following Bobkov et al. (2024), an input image X is encoded into w , edited by a sampled direction from the adaptive editing procedure, and paired with the corresponding edited image $\hat{X}$. This trains H to edit feature tensors according to the category-irrelevant attributes learned in Phase 1. To prevent the Feature Editor from relying only on synthetic editing pairs, it is also trained on the classical inversion task with $\Delta = 0$.

The editing loss is defined in Equation 25:

$$L_{\text{edit}}(X, \hat{X}) = \lambda_{l_2} \|X - \hat{X}\|_2 + \lambda_{\text{lips}} \mathcal{L}_{\text{lips}}(X, \hat{X}). \quad (25)$$

The inversion loss uses the same reconstruction and perceptual terms, as shown in Equation 26:

$$L_{\text{inv}}(X, \hat{X}) = \lambda_{l_2} \|X - \hat{X}\|_2 + \lambda_{\text{lips}} \mathcal{L}_{\text{lips}}(X, \hat{X}). \quad (26)$$

Equations 25 and 26 use L_2 reconstruction and LPIPS perceptual losses to preserve both pixel-level structure and perceptual similarity. The total Feature Editor loss is defined in Equation 27:

$$L_{\text{fe}} = L_{\text{edit}}(X, \hat{X}) + L_{\text{inv}}(X, \hat{X}). \quad (27)$$

In Equation 27, the editing term teaches H to respond to nonzero edit directions, while the inversion term with $\Delta = 0$ stabilizes feature reconstruction. The Feature Editor is trained using the Adam optimizer with a fixed learning rate of 1×10^{-4} for 15,000 iterations on a single

V100 16 GB vGPU. Mixed precision is enabled to improve memory efficiency because the phase requires multiple generator passes. The loss weights are $\lambda_{l_2} = 10$ and $\lambda_{\text{lips}} = 0.8$, with an additional LPIPS scaling factor of 0.5. The sparse dictionary A is initialized with a fixed length of 100.

3.4 Evaluation Protocol

3.4.1 Evaluation Metrics

The models are evaluated using quantitative metrics for generated-image distribution alignment and perceptual variation under a one-shot test-time setup. The same implementations and evaluation settings are used for AGE, SAGE, and FeatureSAGE.

1. Fréchet Inception Distance (FID)

Fréchet Inception Distance (FID) is used to evaluate how closely the distribution of generated images matches the distribution of real images (Heusel et al., 2017; Wang et al., 2023). Instead of comparing images directly at the pixel level, FID computes statistics from a high-level feature space extracted using a pretrained Inception-v3 network. The metric models the real and generated image features as multivariate Gaussian distributions and computes the distance between them. Equation 28 defines the FID score:

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}). \quad (28)$$

In Equation 28, μ_r and Σ_r are the mean and covariance of Inception features from real images; μ_g and Σ_g are the mean and covariance of Inception features from generated images; $\|\mu_r - \mu_g\|_2^2$ is the squared Euclidean distance between feature means; $\text{Tr}(\cdot)$ is the trace operator; and $(\Sigma_r \Sigma_g)^{1/2}$ is the matrix square root of the covariance product. The theoretical basis of FID is the Fréchet, or 2-Wasserstein, distance between two

Gaussian feature distributions (Heusel et al., 2017). Lower FID values indicate stronger distributional alignment between generated and real images.

2. Learned Perceptual Image Patch Similarity (LPIPS)

Learned Perceptual Image Patch Similarity (LPIPS) compares images using deep feature activations from a pretrained neural network, such as VGG or AlexNet, rather than relying only on pixel-level differences (Zhang et al., 2018a). This makes LPIPS useful for evaluating perceptual similarity in reconstruction tasks and perceptual diversity when measured among generated outputs. Equation 29 defines the LPIPS score between images x and x' :

$$\text{LPIPS}(x, x') = \sum_l \frac{1}{H_l W_l} \sum_{h=1}^{H_l} \sum_{w=1}^{W_l} \|\omega_l \odot (\hat{y}_{hw}^l(x) - \hat{y}_{hw}^l(x'))\|_2^2. \quad (29)$$

In Equation 29, $\hat{y}_{hw}^l(x)$ is the unit-normalized feature vector of image x at layer l and spatial position (h, w) , ω_l is a learned channel-weight vector, and H_l and W_l are the height and width of the feature map at layer l . For reconstruction similarity, lower LPIPS values indicate closer perceptual similarity. In the results table, LPIPS is reported as a perceptual-diversity measure among generated outputs, so higher values indicate greater perceptual variation when image fidelity is maintained.

3.4.2 Baselines for Comparison

To contextualize the performance of the proposed few-shot image generation model, FeatureSAGE is compared against two editing-based few-shot image generation baselines:

- **AGE** (Ding et al., 2022): Leverages latent-space attribute editing on pretrained GANs to produce novel-class images without retraining by disentangling class-relevant and category-irrelevant features.
- **SAGE** (Ding et al., 2023): Builds on AGE by aggregating support images to create stable latent embeddings and injecting class-consistent diversity through attribute mixing.

3.5 Expected Outcome

Based on the stated objectives, FeatureSAGE is expected to improve the SAGE pipeline by replacing the pSp-centered inversion stage with the StyleFeatureEditor inverter and Feature Editor. The expected improvement is not a change in the underlying StyleGAN2 generator or in the attribute group editing objective; rather, it is an improvement in how input images are represented before editing. By extracting both W^+ latent codes and intermediate feature tensors, the proposed framework is expected to preserve more local detail while retaining the ability to apply category-irrelevant edits.

The expected outcomes are evaluated on the FUNIT Animal Faces and 102 Flowers datasets under the same one-shot protocol used for the baseline methods. FID is used to measure how closely generated images align with real-image distributions, while LPIPS is used to examine perceptual variation among generated outputs. Within this evaluation design, the expected measurable outcomes are:

1. **Lower FID**, indicating improved visual realism and stronger distributional alignment between real and synthetic images.
2. **Comparable or higher LPIPS as a diversity measure**, indicating that improved fidelity does not eliminate perceptual variation among generated samples.

Together, these outcomes test whether the FeatureSAGE modifications improve few-shot image generation while preserving the edit-based structure of the original SAGE framework.

CHAPTER 4 RESULTS AND DISCUSSION

This chapter presents the experimental results of the proposed FeatureSAGE framework and compares its performance with the AGE and SAGE baselines on few-shot image generation tasks. The experiments are conducted on the FUNIT Animal Faces and 102 Flowers datasets under a one-shot setting, where a single reference image is used for each target class during evaluation. All reported quantitative results are evaluated at 1024×1024 resolution. Generated-image quality is measured using Fréchet Inception Distance (FID), and perceptual variation is measured using Learned Perceptual Image Patch Similarity (LPIPS). The comparison isolates the effect of replacing the pSp-centered SAGE inversion stage with the StyleFeatureEditor inverter and Feature Editor while keeping the same baseline task and evaluation setting.

4.1 Quantitative Results

4.1.1 Quantitative Comparison Between FeatureSAGE and Baselines

Resolution	Model	Flowers		Animal Faces	
		FID ↓	LPIPS ↑	FID ↓	LPIPS ↑
1024	AGE (Baseline)	61.57	0.4108	62.13	0.5033
1024	SAGE (Baseline)	54.17	0.4528	60.92	0.4867
1024	FeatureSAGE (Ours)	45.60	0.4209	49.60	0.4811

Table 4.1 Quantitative comparison of AGE, SAGE, and the proposed FeatureSAGE framework across the 102 Flowers and Animal Faces datasets at 1024×1024 resolution. Lower FID indicates better distributional alignment with real images. Higher LPIPS is interpreted here as greater perceptual variation among generated outputs.

4.2 Analysis of Quantitative Results

This section analyzes the quantitative trends in Table 4.1. The comparison focuses on how FeatureSAGE performs against AGE and SAGE across the 102 Flowers and Animal Faces datasets at 1024×1024 resolution using FID and LPIPS.

4.2.1 102 Flowers

On the 102 Flowers dataset, FeatureSAGE obtains an FID of 45.60, corresponding to a 15.8% improvement over SAGE (54.17) and a 25.9% improvement over AGE (61.57). This FID reduction indicates stronger distributional alignment with real flower images in the Inception feature space. For perceptual variation, FeatureSAGE achieves an LPIPS of 0.4209, compared with 0.4528 for SAGE and 0.4108 for AGE. This represents a 7.0% decrease relative to SAGE and a 2.5% increase relative to AGE. The LPIPS values across the three methods range from 0.4108 to 0.4528, showing that perceptual variation remains within a narrow range while FeatureSAGE achieves the best FID.

4.2.2 Animal Faces

On the Animal Faces dataset, FeatureSAGE achieves an FID of 49.60, outperforming SAGE (60.92) by 18.6% and AGE (62.13) by 20.2%. This reduction indicates improved distributional alignment with the real animal-face data. For LPIPS, FeatureSAGE obtains 0.4811, compared with 0.4867 for SAGE and 0.5033 for AGE. This is a 1.2% decrease relative to SAGE and a 4.4% decrease relative to AGE. The LPIPS values remain close across the three methods, ranging from 0.4811 to 0.5033, which indicates that FeatureSAGE improves FID while maintaining comparable perceptual variation.

4.3 Qualitative Evaluation

4.3.1 Visual Comparison Across Models



Figure 4.1 Comparison across AGE, SAGE, and the proposed FeatureSAGE model under the one-shot setting, where a single reference image is used to condition the model during inference.



Figure 4.2 Comparison across AGE, SAGE, and the proposed FeatureSAGE model under the one-shot setting with a second flower input.



Figure 4.3 Comparison across AGE, SAGE, and the proposed FeatureSAGE model on the Animal Faces dataset under the one-shot setting. Each model generates two outputs per input image to demonstrate edit diversity and semantic consistency.



Figure 4.4 Comparison across AGE, SAGE, and the proposed FeatureSAGE model under the one-shot setting with a second dog input image. Each model generates two outputs per input to illustrate diversity and semantic consistency.

4.3.2 Visual Observations

Flowers

Figures 4.1 and 4.2 compare generated flower images from AGE, SAGE, and FeatureSAGE. The visual analysis focuses on color preservation, edge sharpness, contrast, and semantic consistency. In Figure 4.1, FeatureSAGE retains the pink petal color of the input image and produces sharper petal boundaries than the baseline outputs. The generated image also shows stronger contrast, which contributes to a more defined flower structure. However, the model does not fully preserve the combined red and yellow colors of the pistil from the input image, indicating that fine internal flower details may still be difficult to reconstruct consistently.

Figure 4.2 further shows that FeatureSAGE preserves petal-edge clarity and produces visually coherent flower outputs. One generated output presents a slight tilt, demonstrating that the model can introduce appearance variation while retaining the flower category. Compared with AGE, which appears paler in some regions, and SAGE, which improves contrast but remains less sharp in the shown examples, FeatureSAGE produces clearer and more realistic flower structures in the qualitative comparisons.

Animal Faces

Figure 4.3 compares AGE, SAGE, and FeatureSAGE on an Animal Faces input. All models preserve the basic color and fur texture of the input, but the outputs differ in structural consistency and local realism. AGE and SAGE show more visible variation in orientation between their two generated outputs. FeatureSAGE produces more consistent pose and structure while maintaining coherent fur shading and blending. This supports the quantitative observation that FeatureSAGE improves fidelity while preserving usable perceptual variation.

Figure 4.4 shows similar behavior for the second dog input. FeatureSAGE maintains more consistent eye shape and facial structure than the baselines in the displayed results. Its outputs also show clearer fur texture and stronger contrast. AGE produces greater pose diversity in this example, but the added variation is accompanied by reduced visual clarity. These observations are consistent with the LPIPS results, where FeatureSAGE remains close to the baselines in perceptual variation while achieving the lowest FID on the Animal Faces dataset.

4.4 Failure Cases and Limitations

During experimentation and evaluation of the FeatureSAGE framework, two main limitations were observed:

1. **Dependence on a pretrained StyleGAN2 generator.** FeatureSAGE inherits the coverage and biases of the pretrained StyleGAN2 generator. If the generator does not adequately represent certain categories or visual attributes, those gaps may appear in the generated outputs. This constraint limits the framework to domains where an appropriate pretrained StyleGAN2 model is available and sufficiently aligned with the target data.
2. **Computational overhead.** Integrating the StyleFeatureEditor inverter and Feature Editor increases the computational cost relative to the baseline SAGE setup. These components improve reconstruction and feature-level editing, but they require additional generator passes and feature-tensor operations. As a result, the framework requires

more GPU memory and can slow inference or training, especially when scaling to larger datasets or time-sensitive applications.

These limitations indicate that the reported quality improvements involve a trade-off between image fidelity and computational efficiency. They also motivate future work on lighter inversion modules, more efficient feature editing, and evaluation with generator backbones beyond StyleGAN2.

4.5 Chapter Summary

The results show that FeatureSAGE improves distributional image quality over both baselines on the two evaluated datasets. On 102 Flowers, FeatureSAGE achieves an FID of 45.60, improving over SAGE by 15.8% and over AGE by 25.9%. On Animal Faces, it achieves an FID of 49.60, improving over SAGE by 18.6% and over AGE by 20.2%. LPIPS remains comparable to the baselines, with 0.4209 on Flowers and 0.4811 on Animal Faces, indicating that the FID gains do not remove perceptual variation among generated outputs. Qualitative comparisons further show sharper flower structures, stronger petal clarity, and more coherent animal-face texture and shading in the FeatureSAGE outputs, while the limitations indicate increased computational overhead from the added inverter and Feature Editor components.

CHAPTER 5 SUMMARY, CONCLUSION, AND FUTURE WORK

5.1 Summary of Findings

This study developed and evaluated FeatureSAGE, a modification of the SAGE framework that integrates the StyleFeatureEditor inverter and Feature Editor into the few-shot attribute group editing pipeline. The findings are summarized according to the main contributions and evaluation objectives of the study.

1. Integration of the StyleFeatureEditor inverter. FeatureSAGE replaces the pSp-centered inversion stage of SAGE with the StyleFeatureEditor inverter, allowing the framework to represent input images using both W^+ latent codes and intermediate feature tensors. This modification produced lower FID scores than both baselines on the two evaluated datasets. At 1024×1024 resolution, FeatureSAGE obtained an FID of 45.60 on 102 Flowers and 49.60 on Animal Faces.

2. Use of feature-level editing. The Feature Editor modifies intermediate feature representations during the decoder stage, supporting detail preservation while latent-space edits control semantic variation. Qualitative comparisons show sharper petal boundaries, clearer flower structure, and more coherent animal-face texture and shading in the FeatureSAGE outputs compared with the displayed AGE and SAGE outputs.

3. Quantitative improvement over AGE and SAGE. On the 102 Flowers dataset, FeatureSAGE improved FID by 15.8% over SAGE and 25.9% over AGE. On the Animal Faces dataset, FeatureSAGE improved FID by 18.6% over SAGE and 20.2% over AGE. These results show that the proposed inversion and feature-editing changes improved distributional alignment with real images under the one-shot setting.

4. Preservation of perceptual variation. FeatureSAGE maintained LPIPS values close to the baselines while achieving the best FID scores. On 102 Flowers, FeatureSAGE obtained an LPIPS of 0.4209, compared with 0.4528 for SAGE and 0.4108 for AGE. On Animal Faces, FeatureSAGE obtained an LPIPS of 0.4811, compared with 0.4867 for SAGE and 0.5033 for AGE. Thus, the improvement in FID did not remove perceptual variation among generated samples.

5.2 Limitations

The findings are constrained by the scope of the study. First, FeatureSAGE depends on a pretrained StyleGAN2 generator, so its applicability is limited to domains where an appropriate generator is available and sufficiently representative of the target categories. Any coverage gaps or biases in the pretrained generator can influence the generated outputs.

Second, integrating the StyleFeatureEditor inverter and Feature Editor increases computational requirements relative to the baseline SAGE pipeline. The additional feature extraction, feature editing, and generator passes improve reconstruction and output quality, but they also increase GPU memory use and processing time.

Third, the quantitative evaluation is limited to the FUNIT Animal Faces and 102 Flowers datasets under a one-shot setting at 1024×1024 resolution. The conclusions should therefore be interpreted within these datasets, metrics, and experimental conditions.

5.3 Future Work

Future research may focus on reducing the computational overhead introduced by the StyleFeatureEditor inverter and Feature Editor. More efficient inversion or feature-editing modules could preserve the FID improvements reported in this study while lowering GPU memory use and inference time.

Another direction is to evaluate FeatureSAGE with generator backbones beyond StyleGAN2. Since the present study is limited to the StyleGAN2-based SAGE pipeline, testing

other generator architectures would clarify whether the same feature-level editing strategy generalizes beyond the current framework.

Future work may also examine multi-level feature editing. The current implementation uses an intermediate feature tensor for feature injection; extending the approach to multiple feature levels could further test whether local detail preservation and semantic consistency can be improved without sacrificing perceptual variation.

5.4 Conclusion

This study addressed the problem of improving few-shot image generation in the SAGE framework, particularly the limitation caused by relying on pSp-based inversion. The proposed FeatureSAGE framework integrates the StyleFeatureEditor inverter and Feature Editor so that images are represented not only by W^+ latent codes but also by intermediate feature tensors used during synthesis.

The results answer the research problem affirmatively within the scope of the evaluated datasets and metrics. Integrating the StyleFeatureEditor inverter and Feature Editor improved generated-image distribution alignment under the one-shot setting. FeatureSAGE achieved the lowest FID in all reported comparisons: 45.60 on 102 Flowers and 49.60 on Animal Faces. These correspond to FID improvements of 15.8% and 18.6% over SAGE, and 25.9% and 20.2% over AGE, respectively. At the same time, LPIPS remained comparable to the baselines, with 0.4209 on Flowers and 0.4811 on Animal Faces, showing that the fidelity gains did not eliminate perceptual variation.

The findings address the research questions as follows. First, the integration of the StyleFeatureEditor inverter and Feature Editor improved the inversion-and-editing pipeline as reflected by lower FID scores and sharper qualitative outputs. Second, the modifications influenced generated-image quality while maintaining comparable LPIPS values, indicating that consistency and diversity remained balanced in the reported one-shot setting. Third, FeatureSAGE retained the SAGE attribute group editing structure while adding feature-level

editing, supporting reliable category-preserving edits in the displayed qualitative results. Fourth, compared with AGE and SAGE, FeatureSAGE achieved the strongest overall quantitative performance on FID across both datasets while remaining close to the baselines in LPIPS.

The study contributes four main findings: it integrates StyleFeatureEditor inversion into SAGE, adds feature-level editing through the Feature Editor, demonstrates consistent FID improvement over AGE and SAGE on both datasets, and shows through LPIPS and qualitative comparisons that perceptual variation and visual coherence are maintained. Therefore, FeatureSAGE provides a stronger few-shot attribute group editing framework for the tested Animal Faces and 102 Flowers settings, while its dependence on StyleGAN2 and its higher computational cost remain important limitations for future work.

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